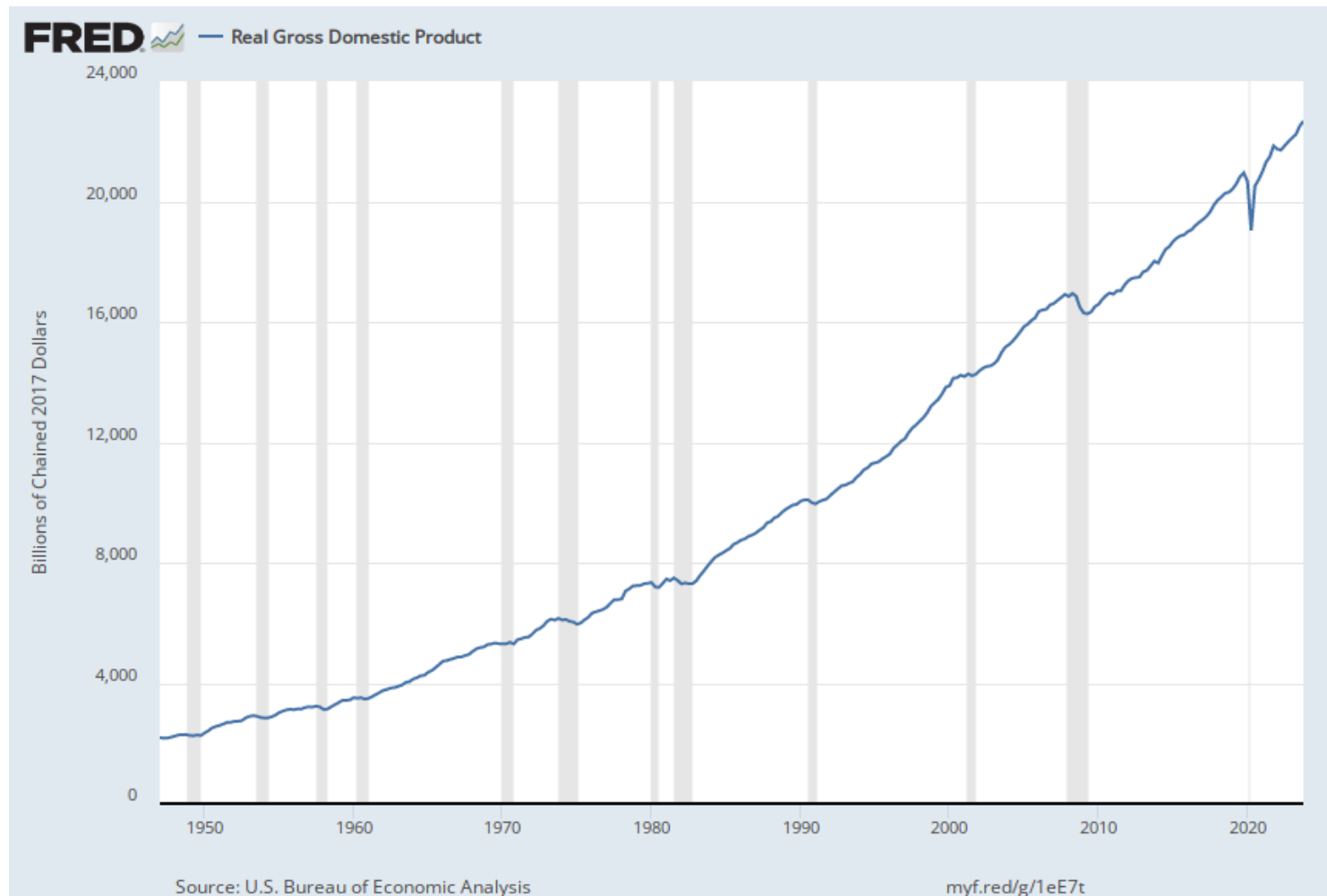


Chapter 1

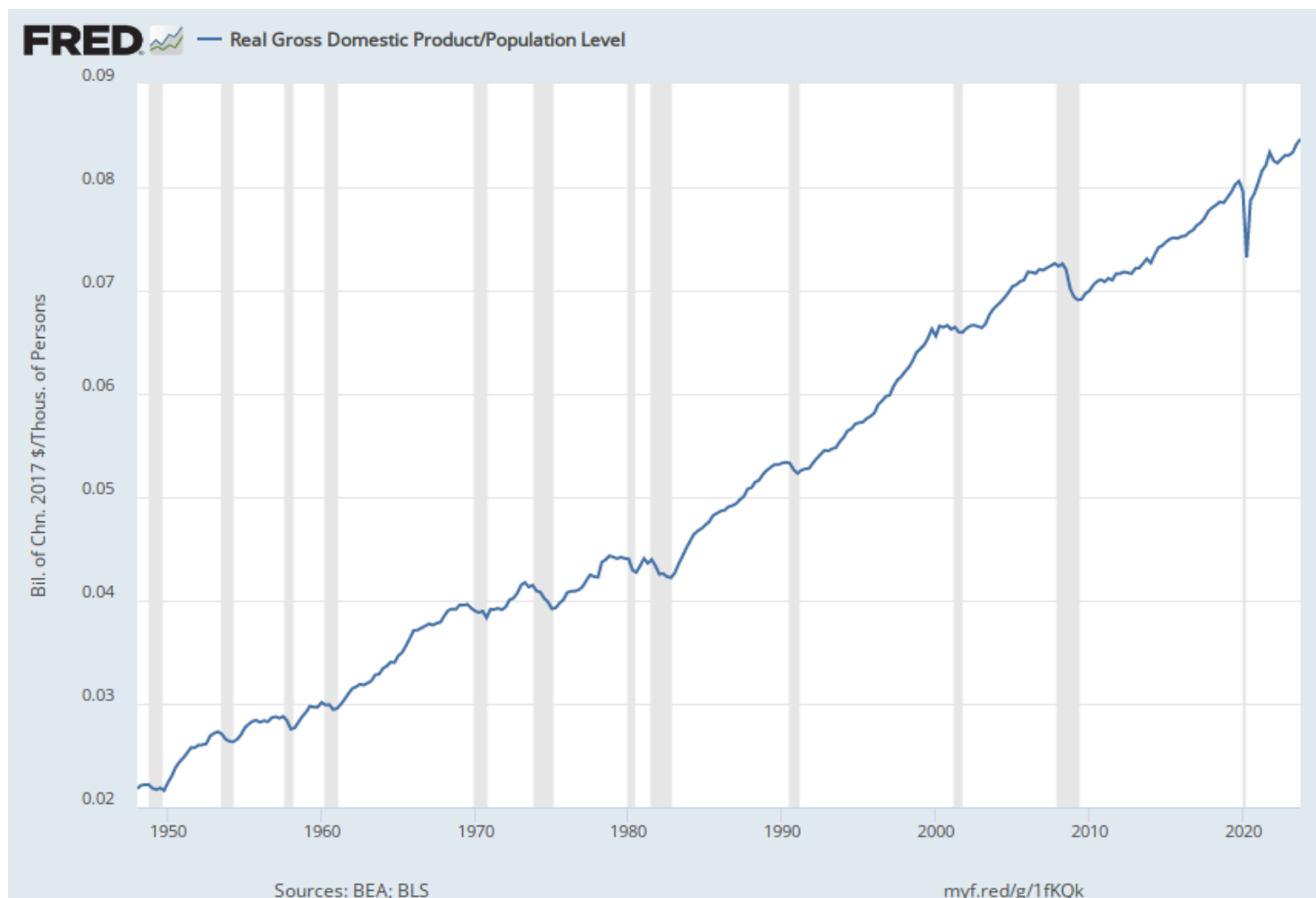
Introduction

GDP in the US since 1948



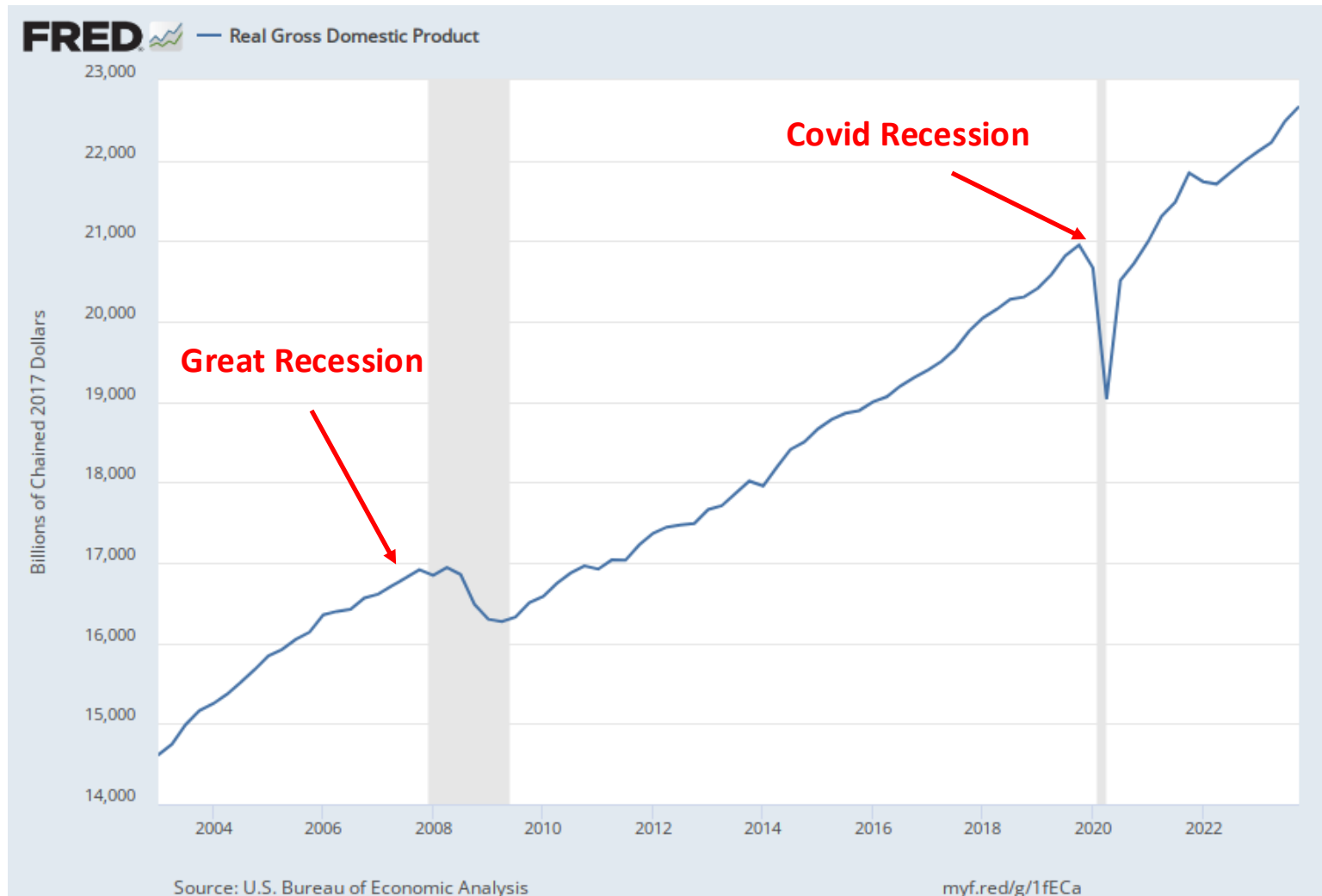
Output grows over time
But also exhibits cyclical fluctuations around its long-term trend

GDP per capita (=GDP/POP) in the US since 1948



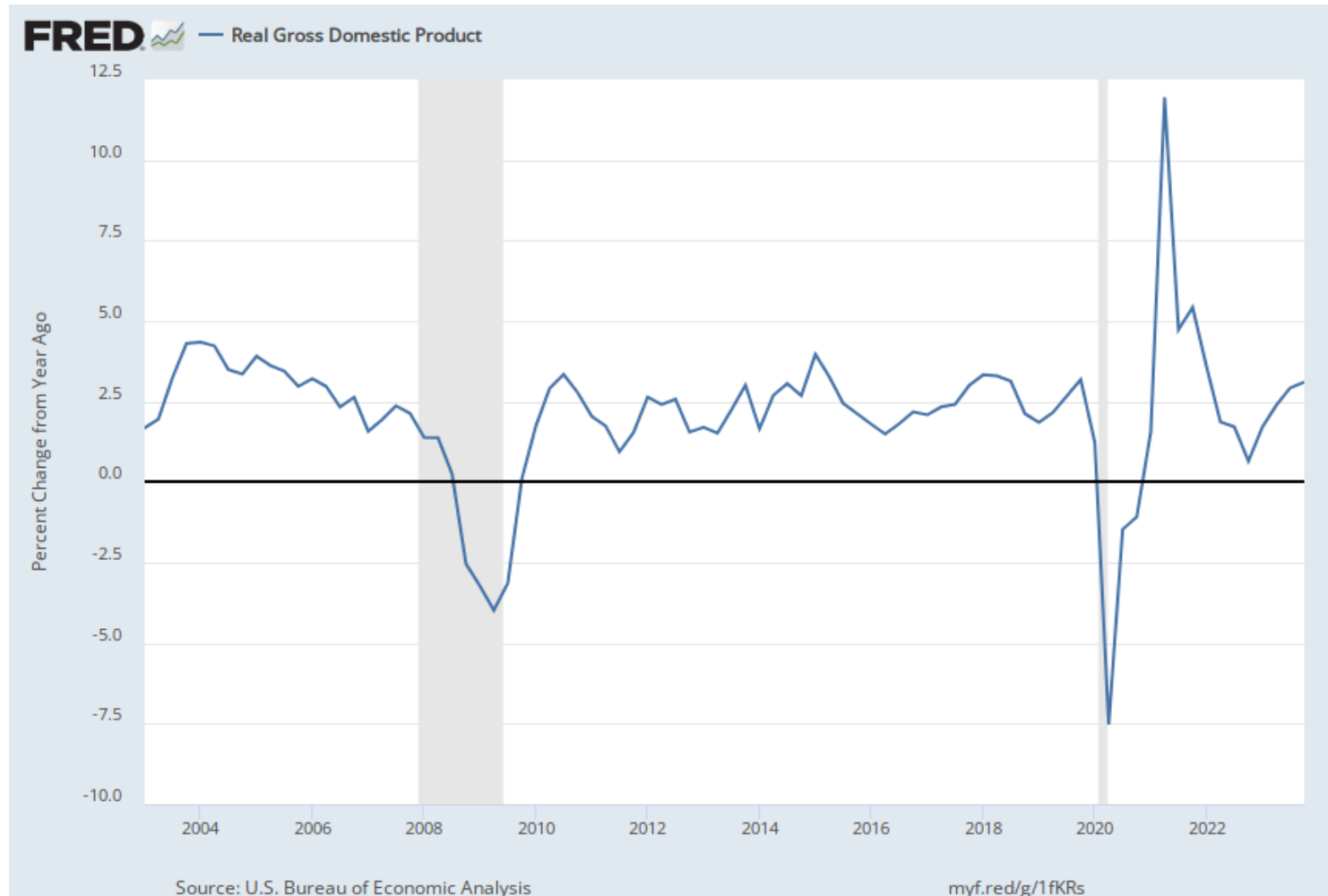
Output per capita also grows over time, as countries become richer

GDP in the US since 2003



When output drops below trend, the economy is in a recession
Economies cycle between recessions and expansions

GDP growth in the US since 2003



One way to de-trend is to look at growth rates

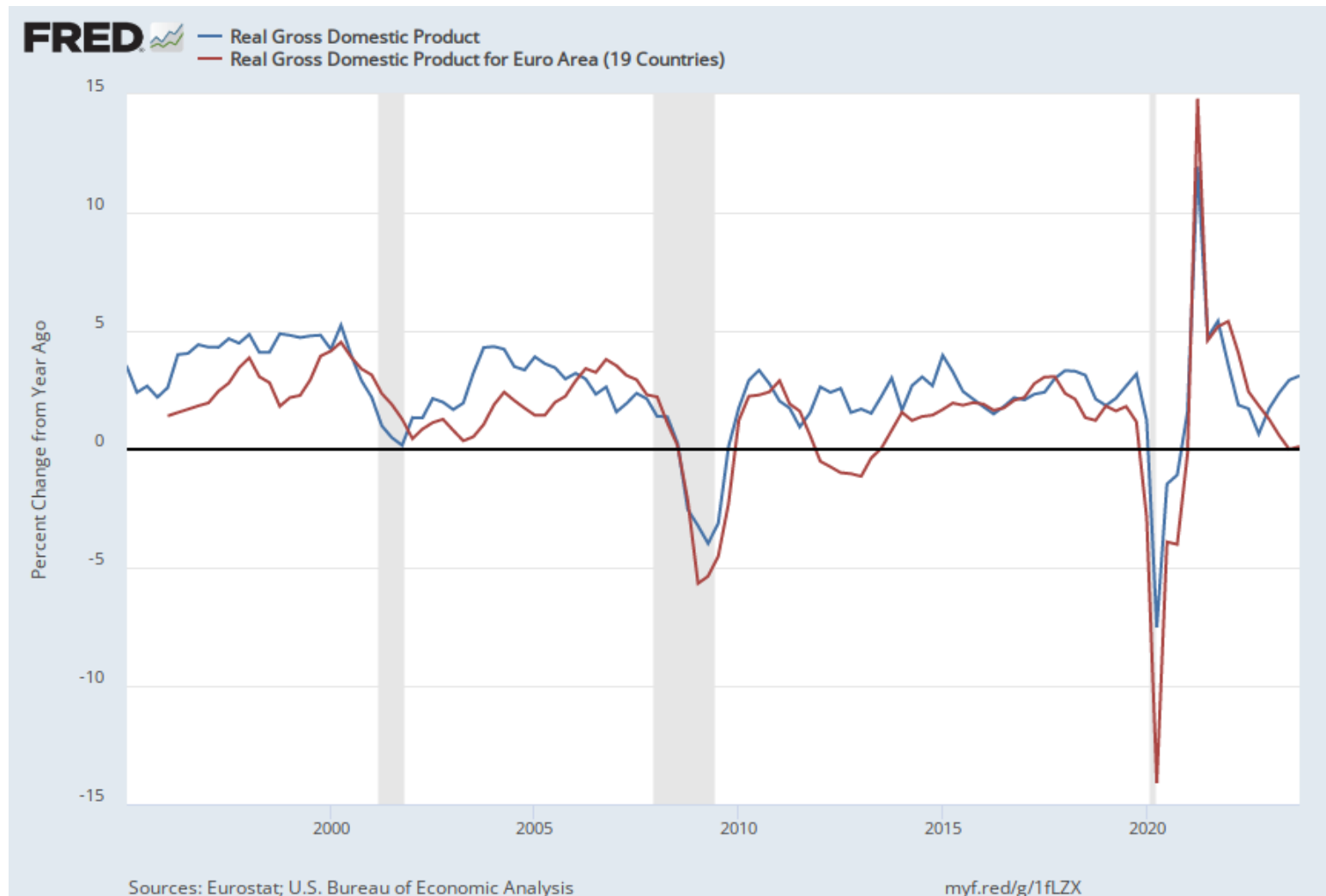
GDP: level versus growth rate

- Real GDP level in period t , Y_t
- Real GDP growth (or rate of change between $t-1$ and t):

$$(Y_t - Y_{t-1}) / Y_{t-1}$$

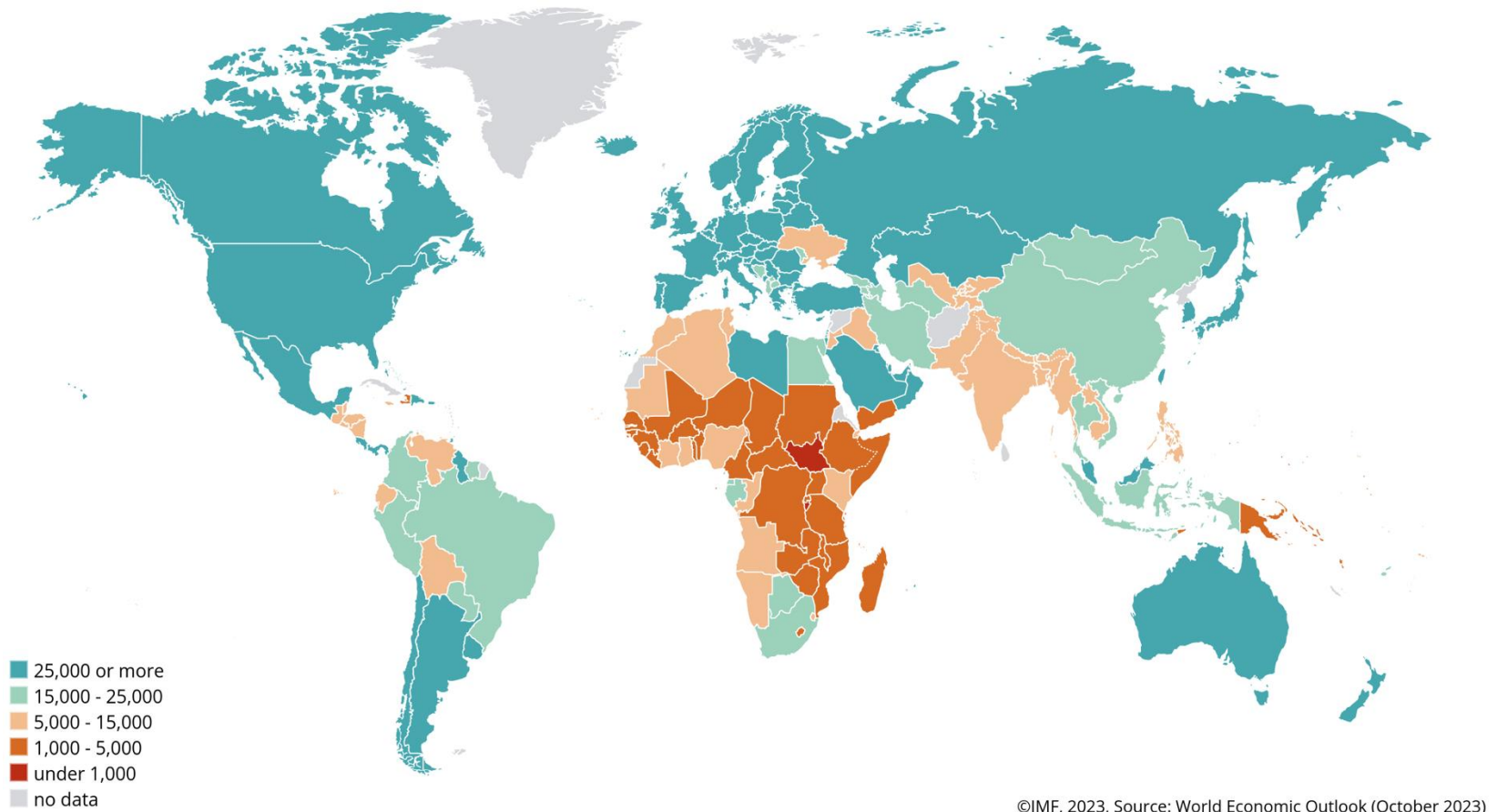
- Periods of **positive GDP growth** are called **expansions**
- Periods of **negative GDP growth** (at least 2 consecutive quarters) are called **recessions**

GDP growth in the US and the EA since 2003



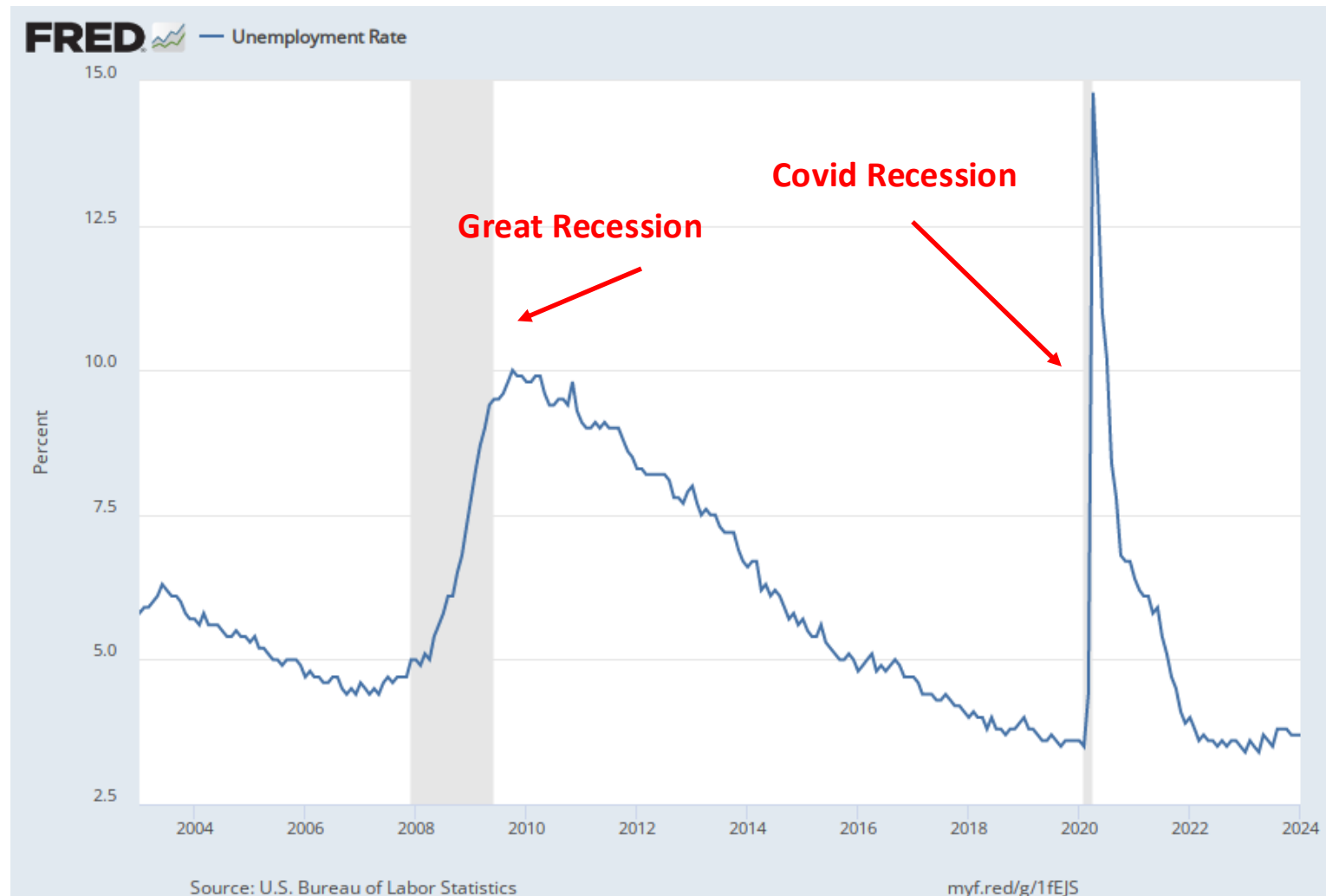
There is a high degree of global comovement

GDP per capita in the world (PPP) in 2024



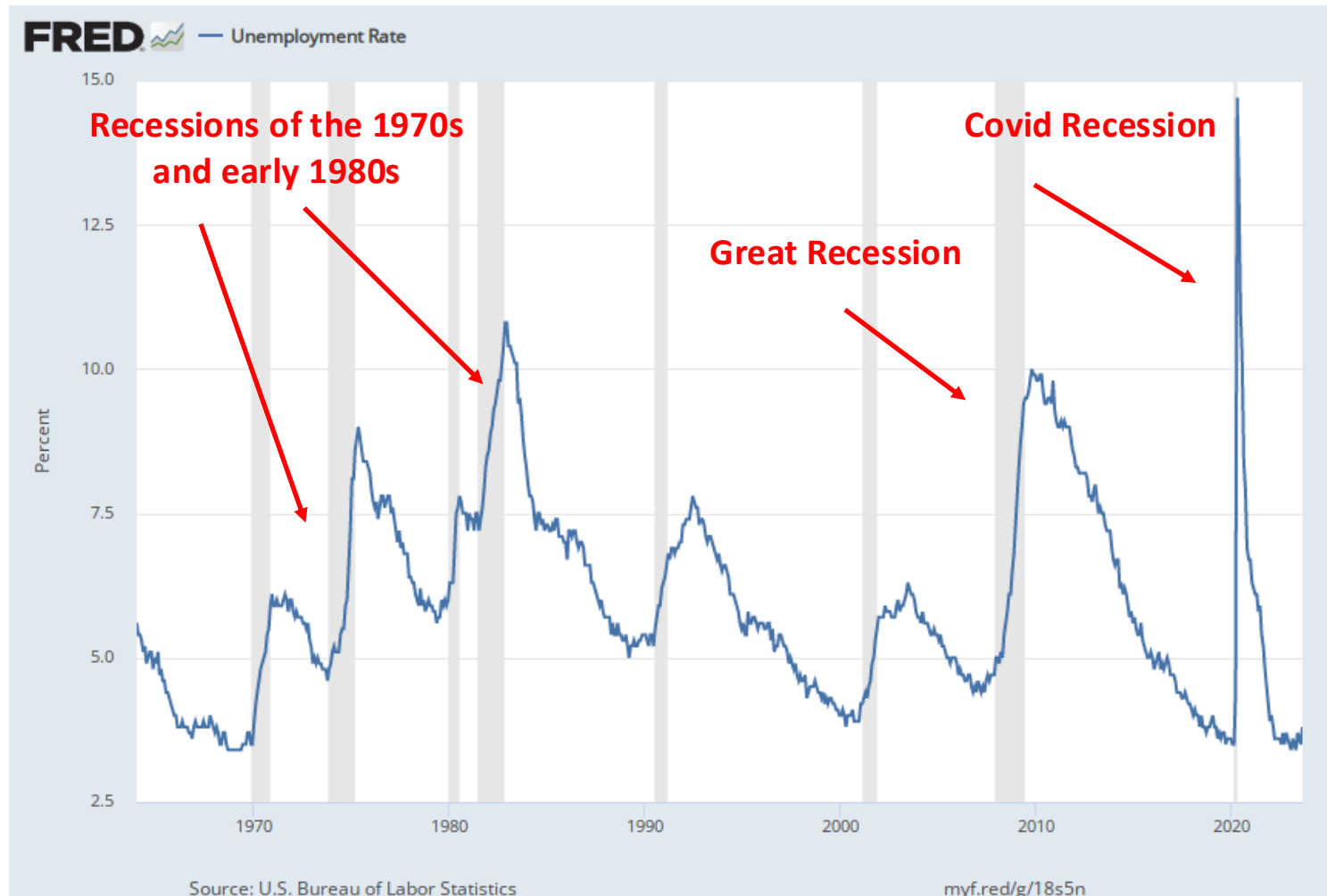
The US GDP per capita is 83,000 US dollars. Italy GDP per capita is 56,000 US dollars.
The Republic of South Sudan GDP per capital is 493 US dollars.

Unemployment rate in the US since 2003



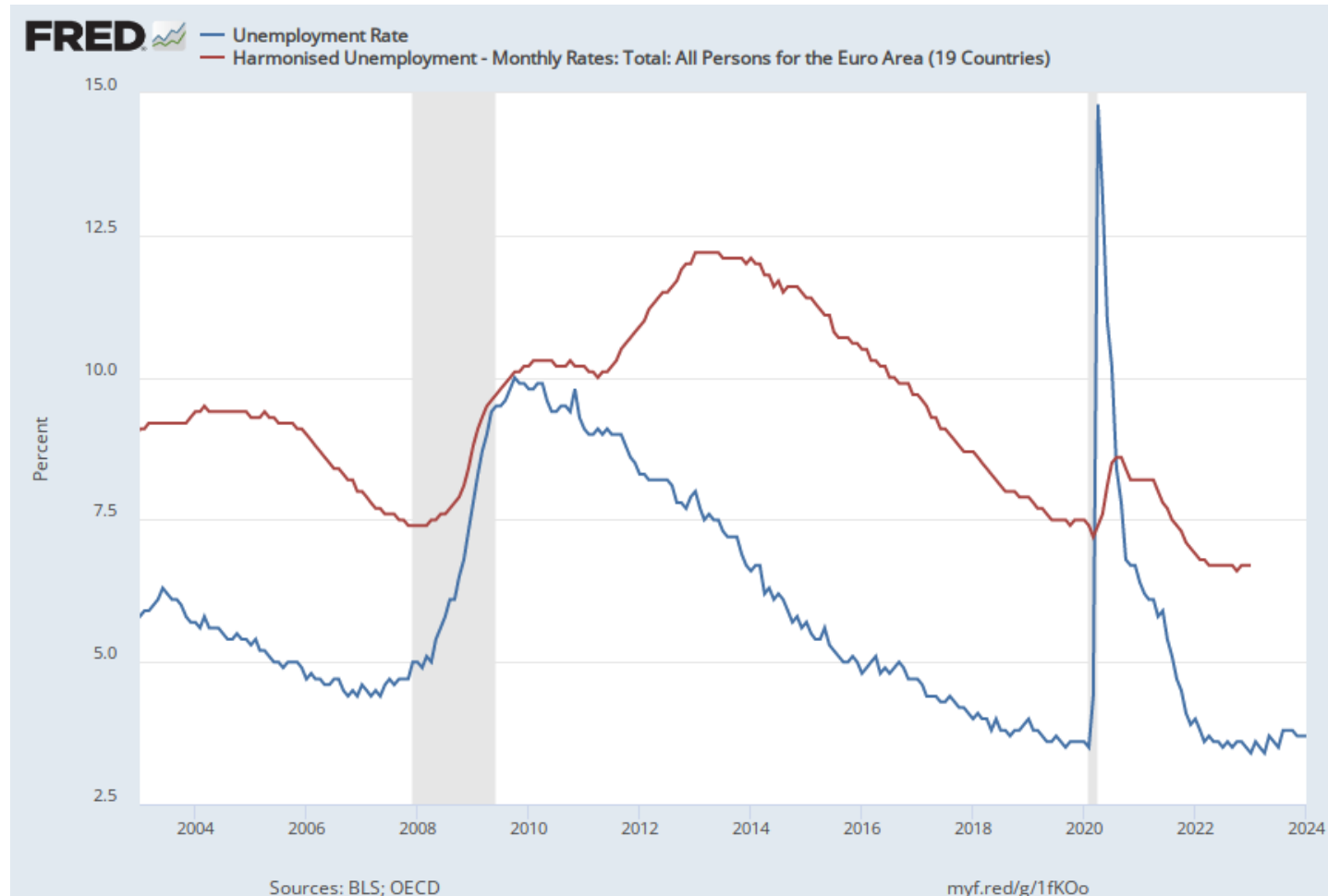
Unemployment always rises in recessions and falls in expansions

Unemployment rate in the US since 1964



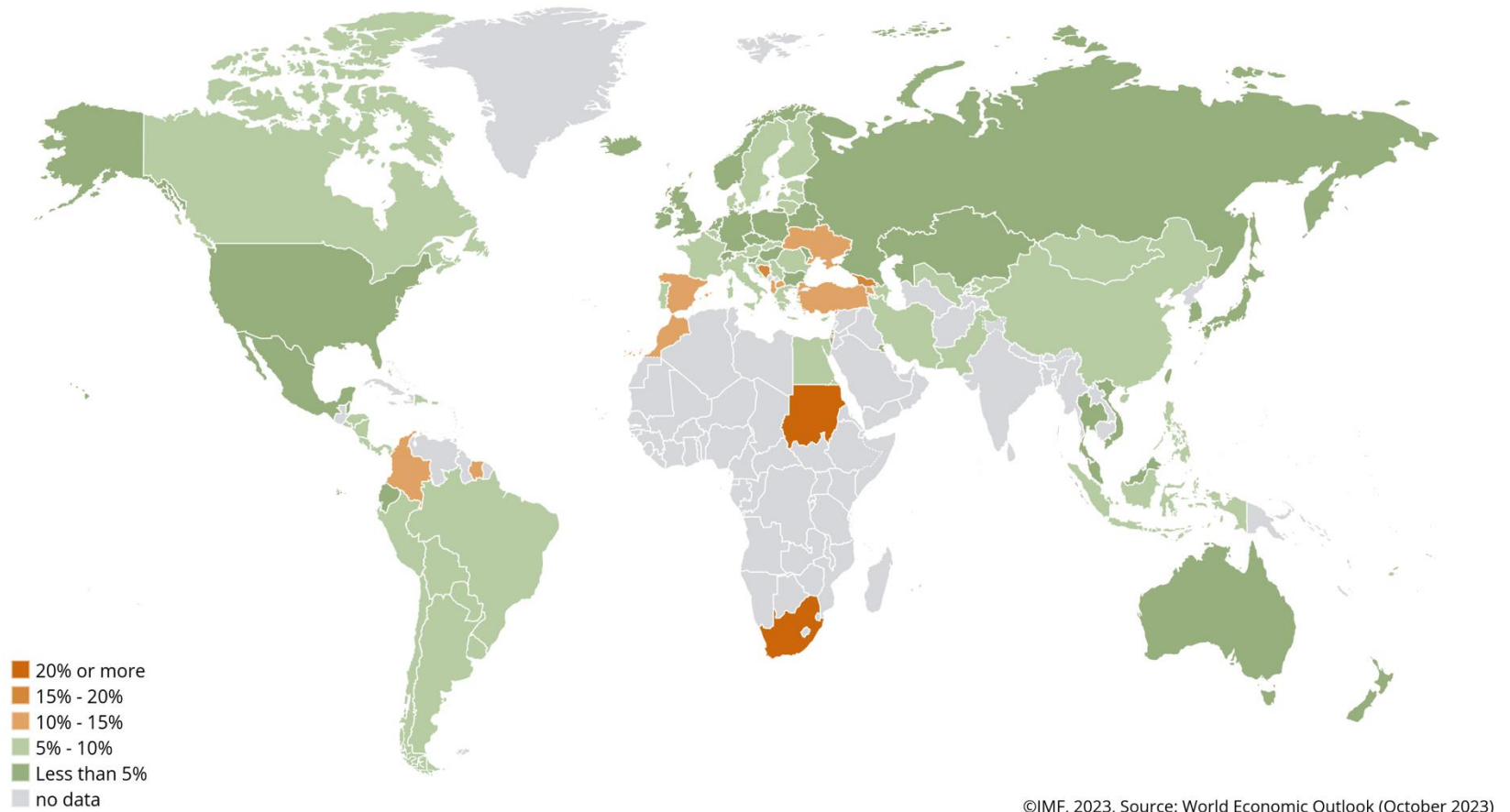
Unemployment always rises in recessions and falls in expansions

Unemployment rate in the US and EA since 2003



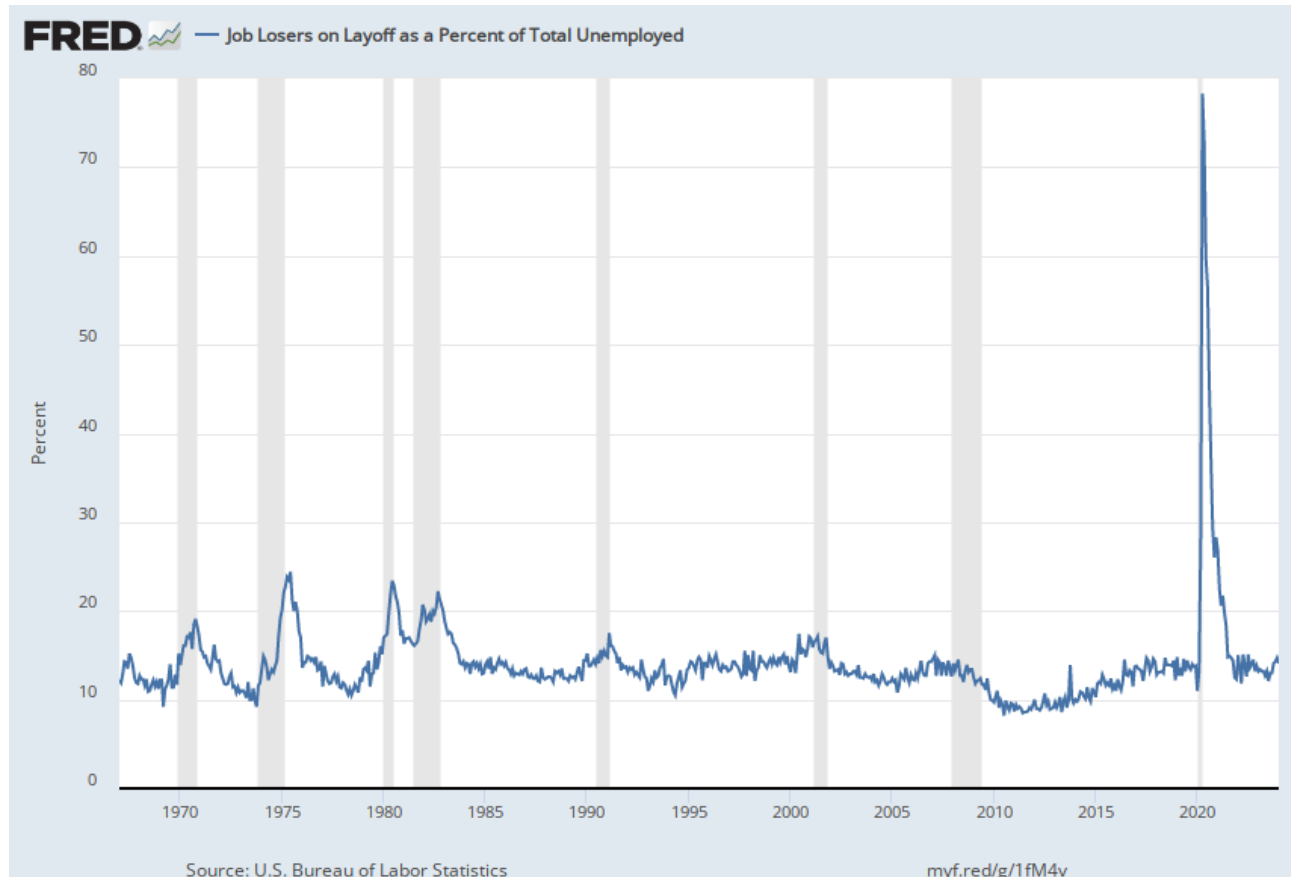
Country-specific labor market institutions are one explanations for different unemployment patterns in the US and the EA

Unemployment rates in the world in 2024



Large differences within advanced economies and Europe:
US: 3.8%. Italy: 8%. Germany: 3.3%. Spain: 11.3%.

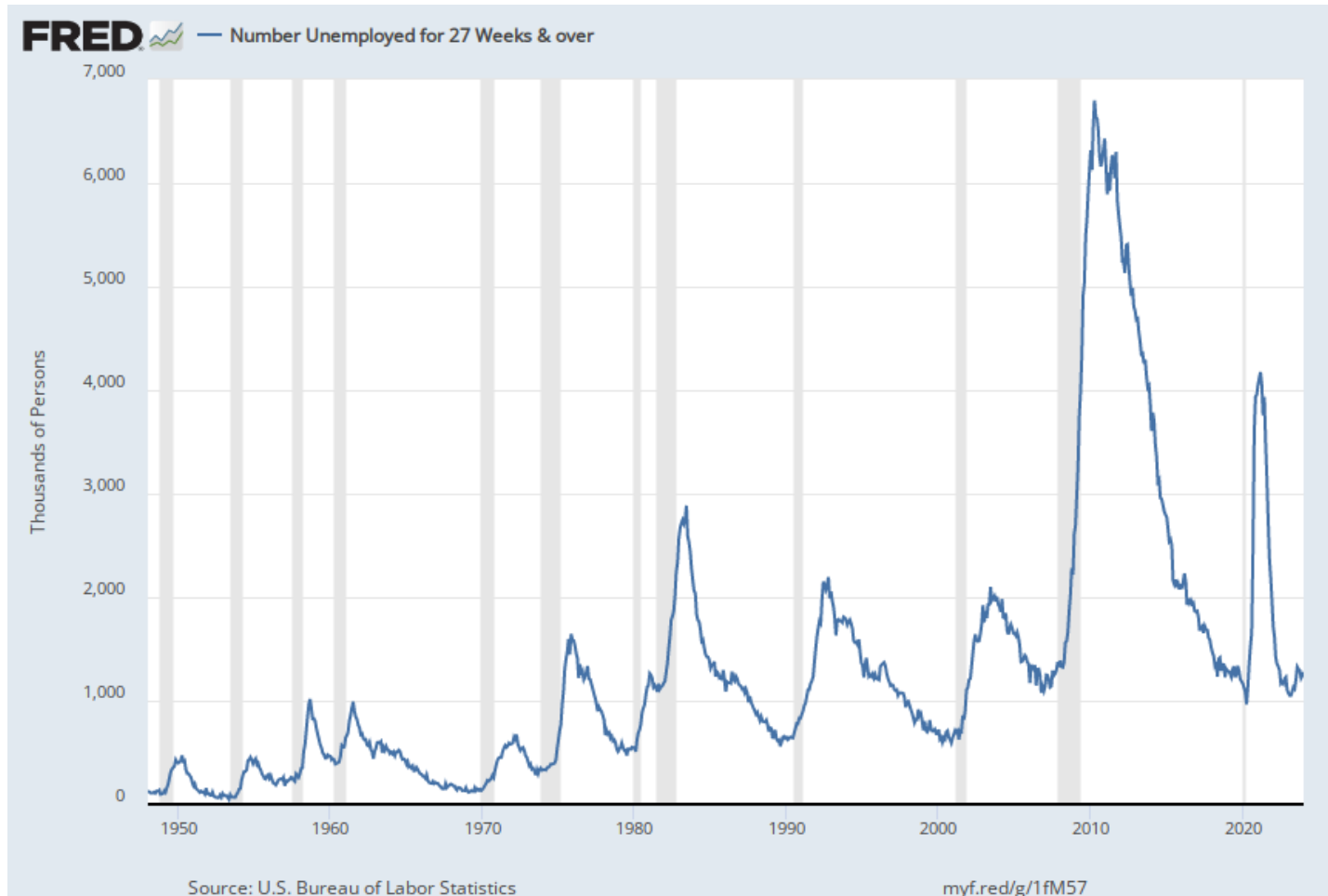
Temporary-Layoff Unemployment in the US since 1967



Total unemployment masks significant heterogeneity within countries as well:

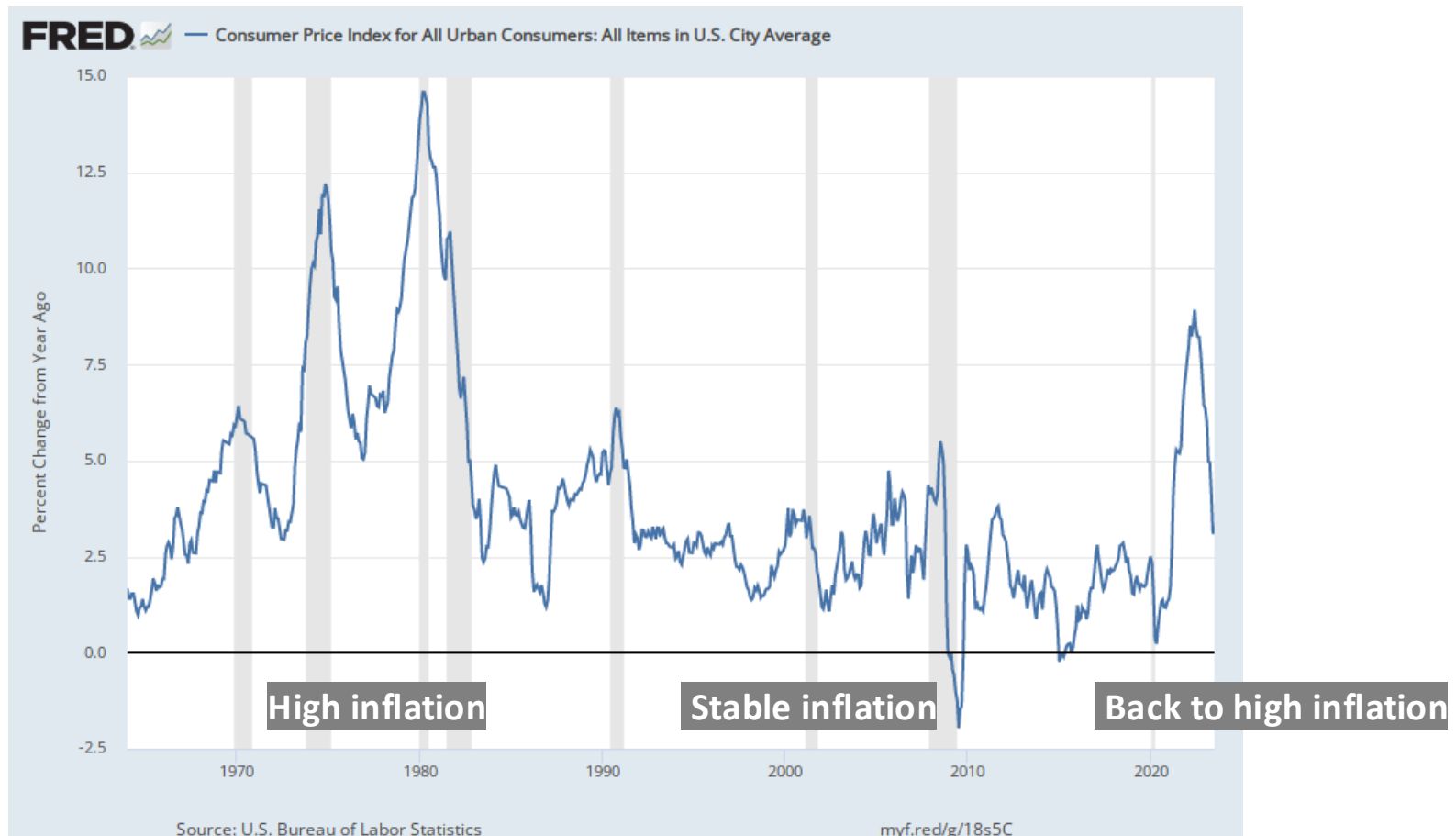
1. Some workers are only temporarily laid-off and expect to return to work quickly

Long-Term Unemployment in the US since 2003



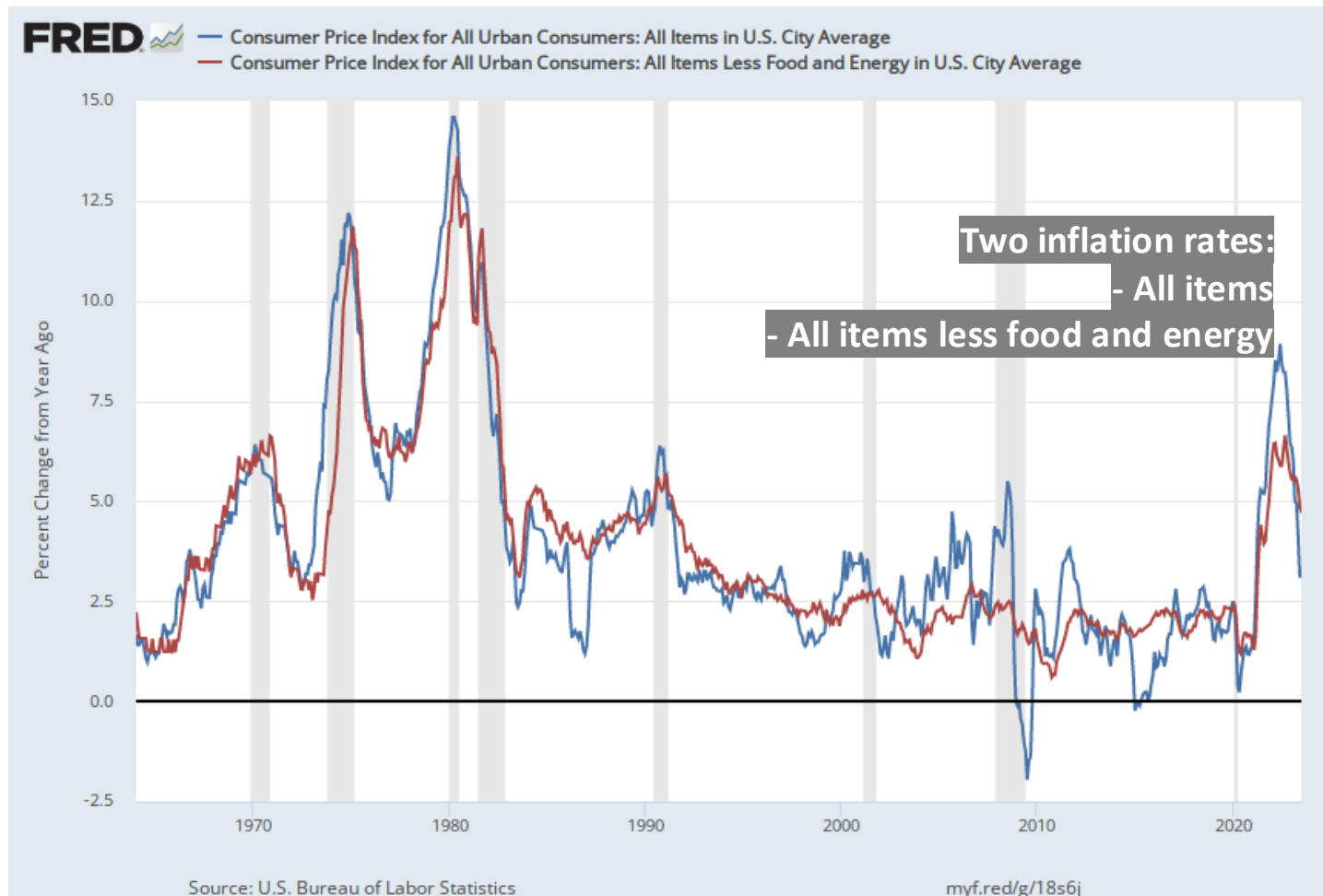
2. Other workers remain unemployed for a very long time

Inflation rate in the US since 1964



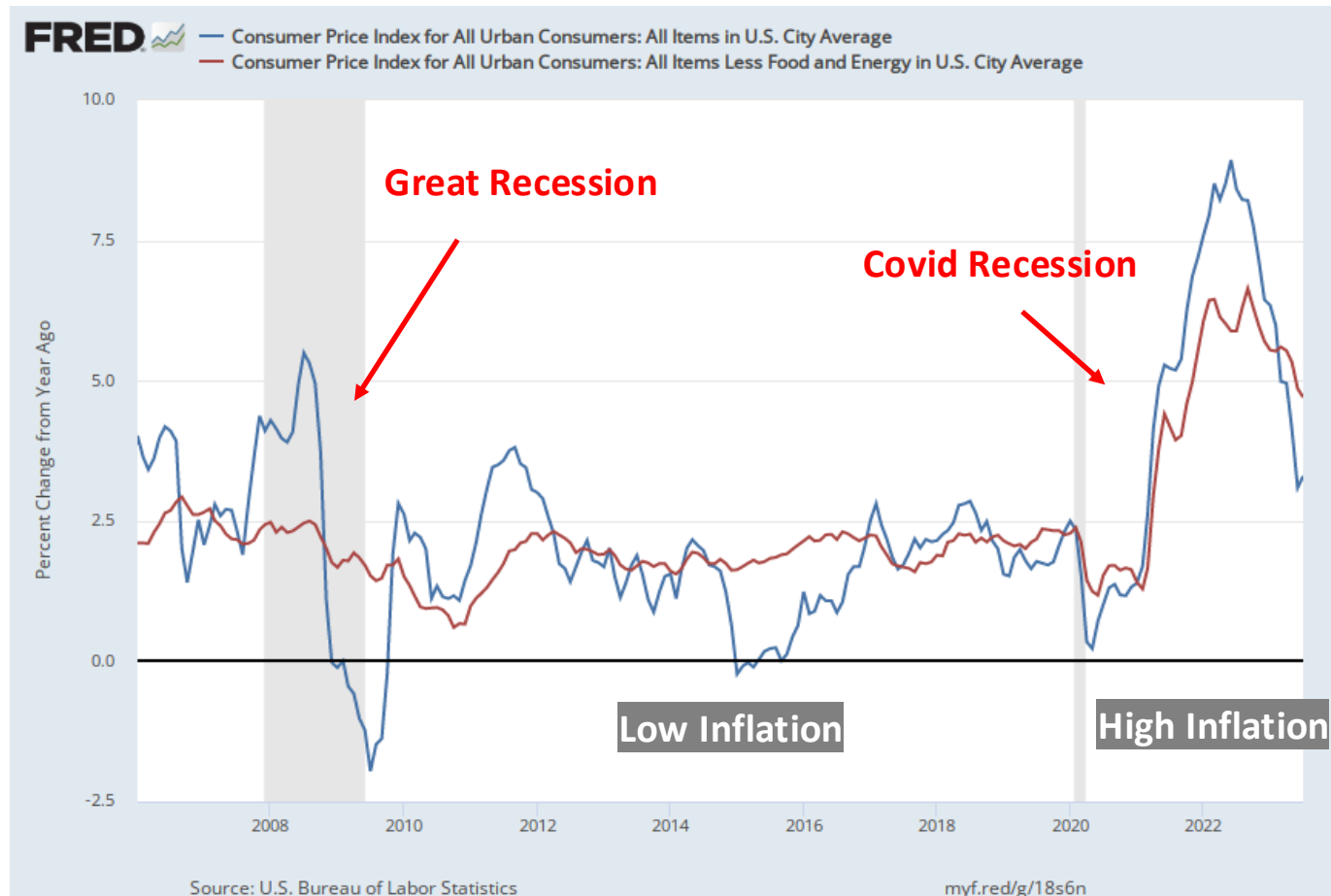
In general, inflation drops in recessions and raises in recoveries
Inflation was high in the 70s, stable around 2% for more than 2 decades and high again today

Two inflation rates in the US since 1964



Inflation «less food and energy» much less volatile

Two inflation rates in the US since 2006



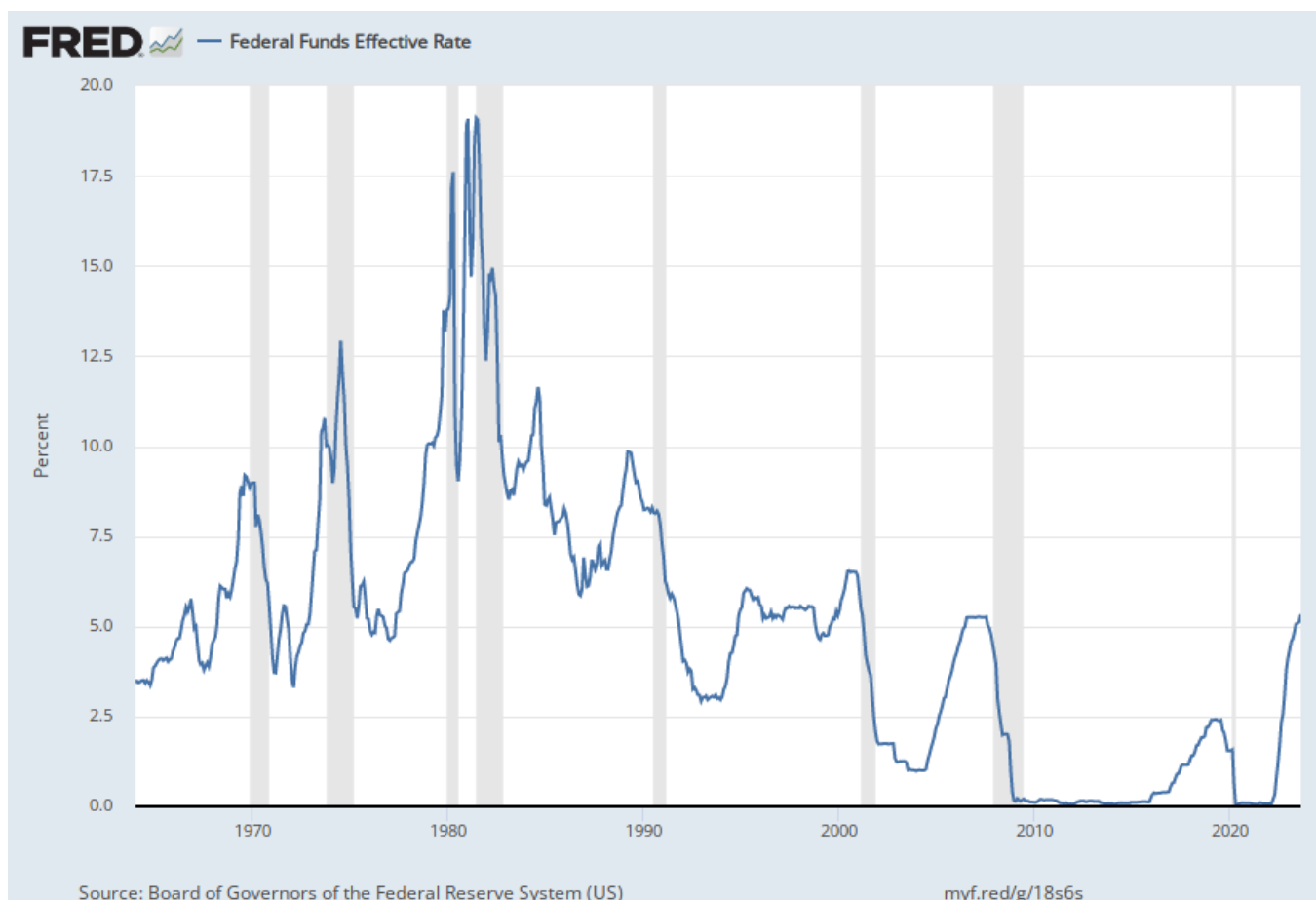
Fed has struggled with too low inflation in the 2010s and faces too high inflation now

Inflation rate in the EA since 1997



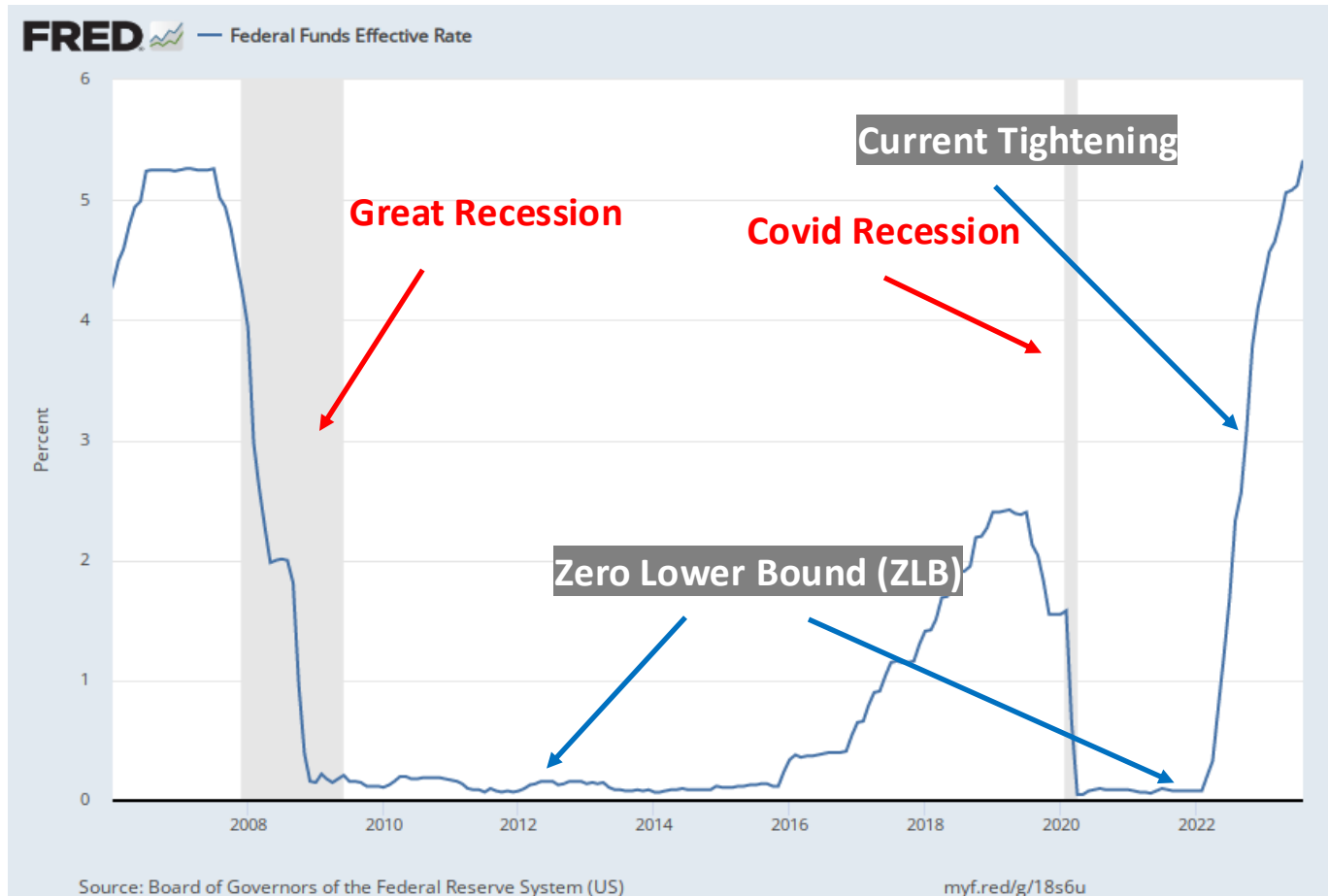
ECB as well has struggled with too low inflation in the 2010s and faces too high inflation now

Monetary policy rate in the US since 1964



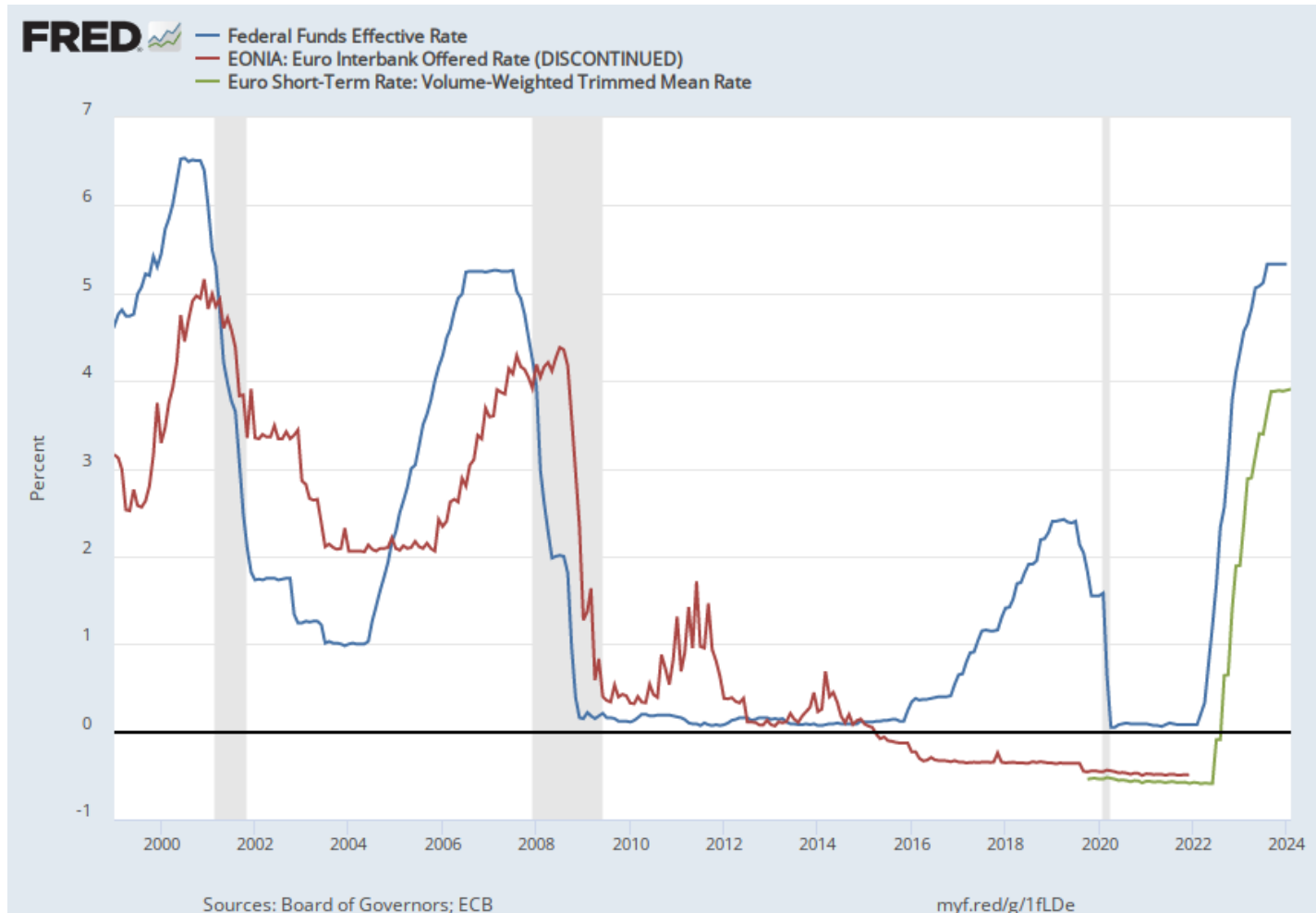
Countercyclical monetary policy:
policy rate is reduced in recessions and raised in expansions

Monetary policy rate in the US since 2006



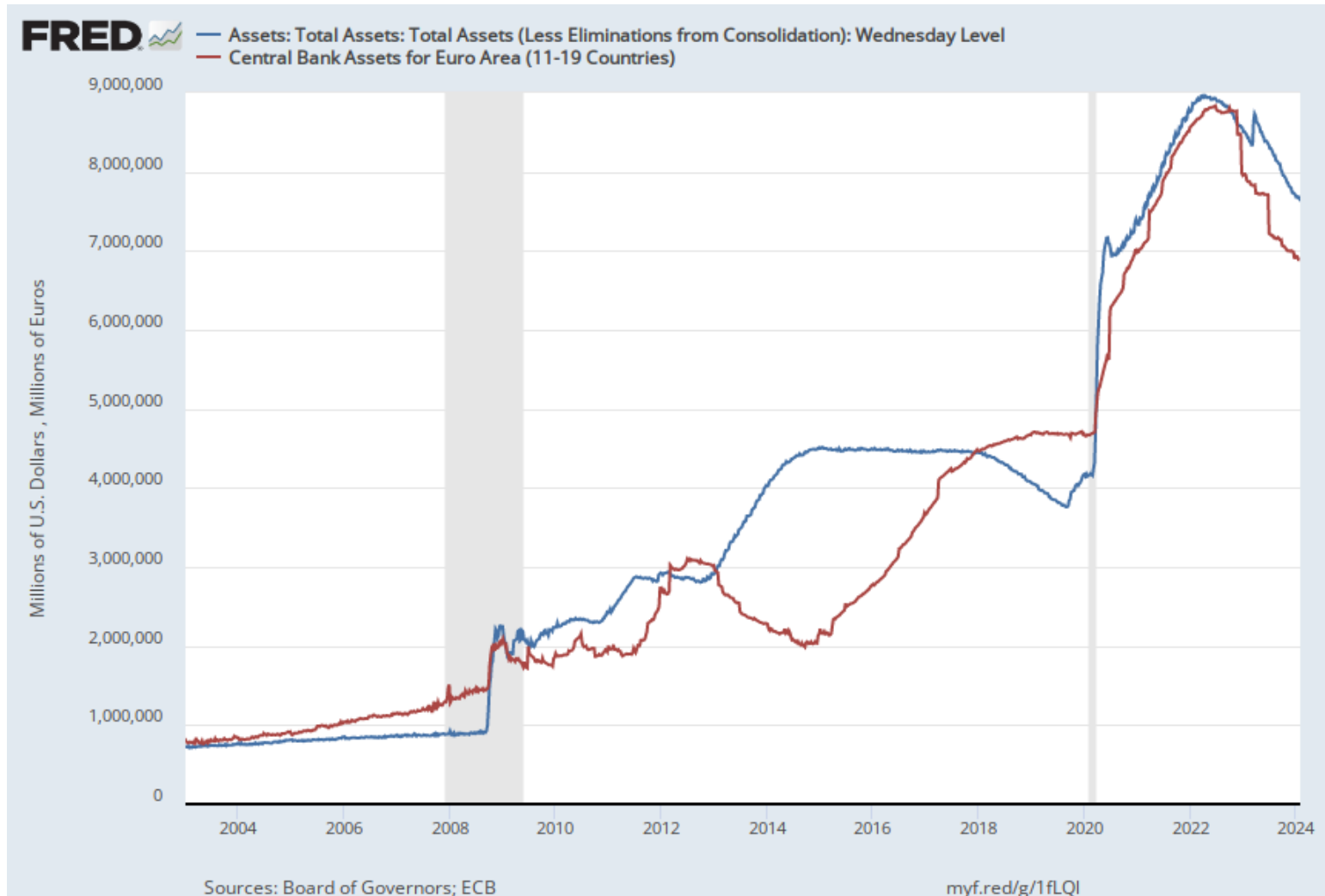
During both the Great Recession and the Covid recession, the FFR reached the ZLB
Since March 2022, the Fed has been tightening monetary policy

Monetary policy rates in the US and the EA since 2003



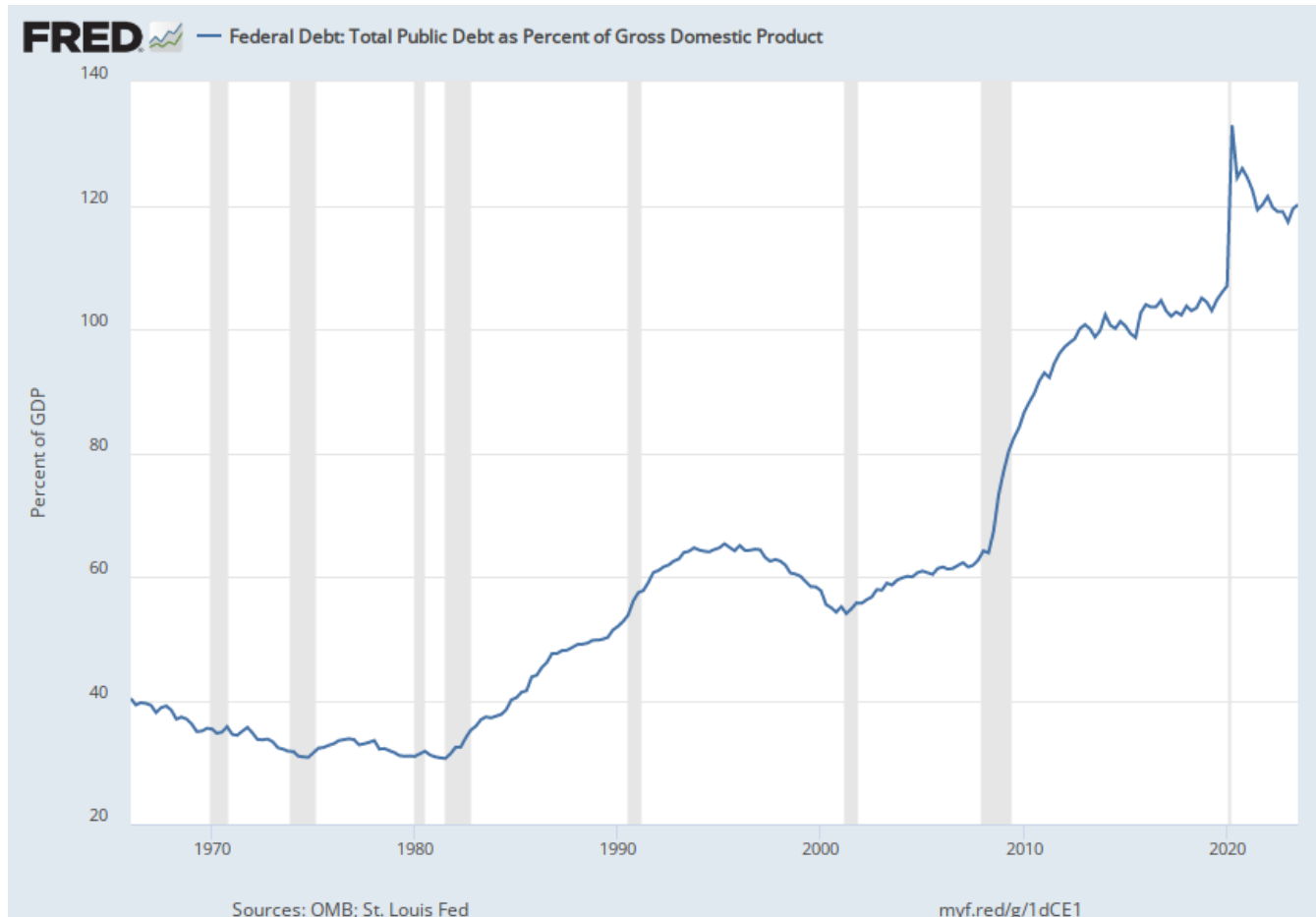
ECB similarly conducted an expansionary monetary policy during the Great Recession and the Covid recession, and a contractionary monetary policy since 2022

Fed and ECB total assets since 2003



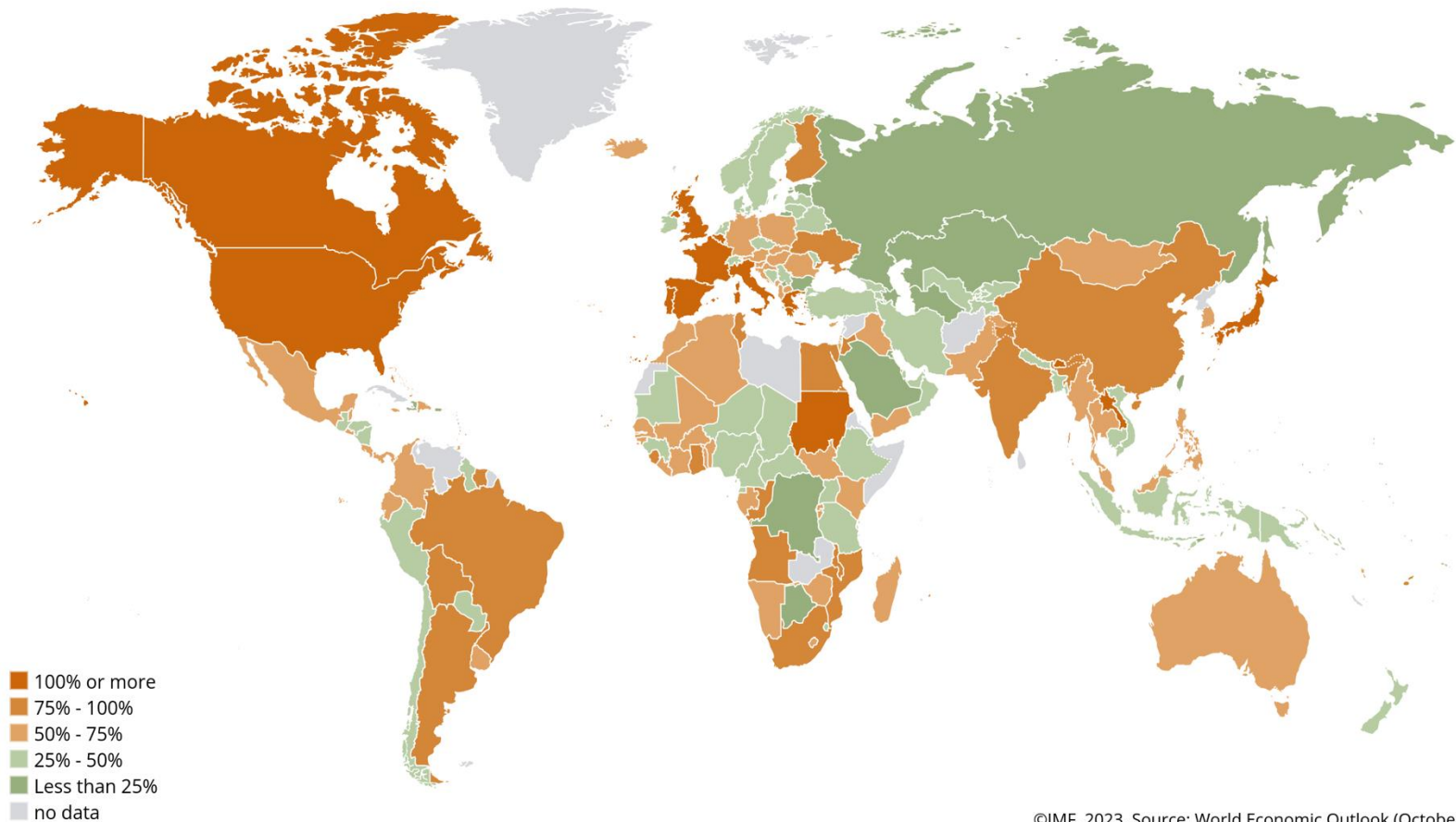
Quantitative easing: increases in the central bank balance sheet

Public debt (% of GDP) in the US since 1966



Expansionary fiscal policies in Great Recession and Covid recession left the US with massive levels of debt

Public debt (% of GDP) in the world in 2024



©IMF, 2023, Source: World Economic Outlook (October 2023)

The massive increase in public debt in recent decades is shared by most advanced economies
US: 126.9%. Italy: 143.2%. Japan: 251.9.3%.

Chapter 2

Aggregate Statistics

Three aggregate statistics

When macroeconomists study an economy, they first look at 3 variables:

- Output
- Unemployment rate
- Inflation rate

Aggregate output

- National income and product accounts

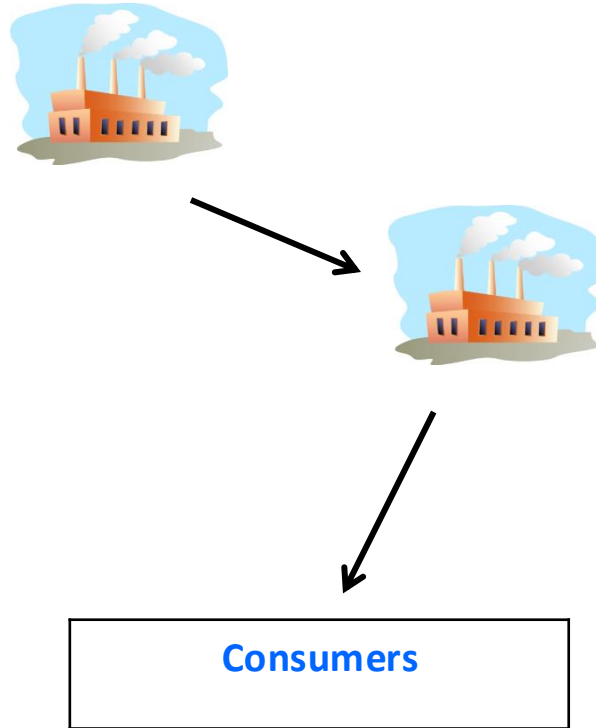
Accounting system used to measure aggregate economic activity

Developed at the end of World War II

- Measure of aggregate output in the national income accounts

Gross Domestic Product (GDP)

Example A



Steel company (Firm 1)

Revenues from sales	€100
Expenses	€80
Wages	€80
Profits	€20

Car company (Firm 2)

Revenues from sales	€200
Expenses	€170
Wages	€70
Steel purchases	€100
Profits	€30

GDP = ?

GDP: Definition 1 (Production side)

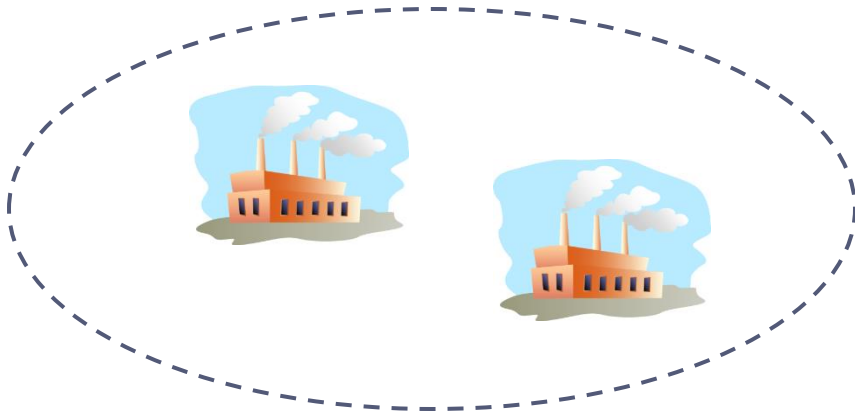
There are 3 ways of defining GDP:

1. GDP is the *value of final goods and services produced in the economy during a given period*
 - A **final good** is a good that is intended for final consumption
 - An **intermediate good** is a good used in the production of another good
- In example A
- Cars are final goods (€200)
 - Steel is an intermediate good (€100)

GDP = €200

Example B

Firm 1 merges with Firm 2



Steel and car company (Firm 1+2)

Revenues from sales of cars	€200
Expenses	€150
Wages	€80+€70=€150
Profits	€50

Consumers

GDP = ?

→ In example B

No sale or record of intermediate goods, steel

Only production of final goods, cars (€200)

GDP = €200

GDP: Definition 2 (Production side)

2. GDP is the *sum of value added* in the economy during a given period

Value added by a firm is the value of its *production minus* the value of the *intermediate goods* used in production

→ In example A

Value added by firm 1 is €100

Value added by firm 2 is €200-€100 =€100

$$\text{GDP} = \text{€100} + \text{€100} = \text{€200}$$

→ In example B, value added by the single firm is €200

$$\text{GDP} = \text{€200}$$

GDP: Definition 3 (Income side)

3. GDP is the *sum of incomes* in the economy during a given period

Value added is used to remunerate

- Workers → labor income
- Firms → capital income or profit income
- Government → indirect taxes

(such as VAT or a sale tax), it is a tax collected by an intermediary (such as a retail store) from the subject who bears the economic burden of the tax (such as the consumer)

→ In examples A and B

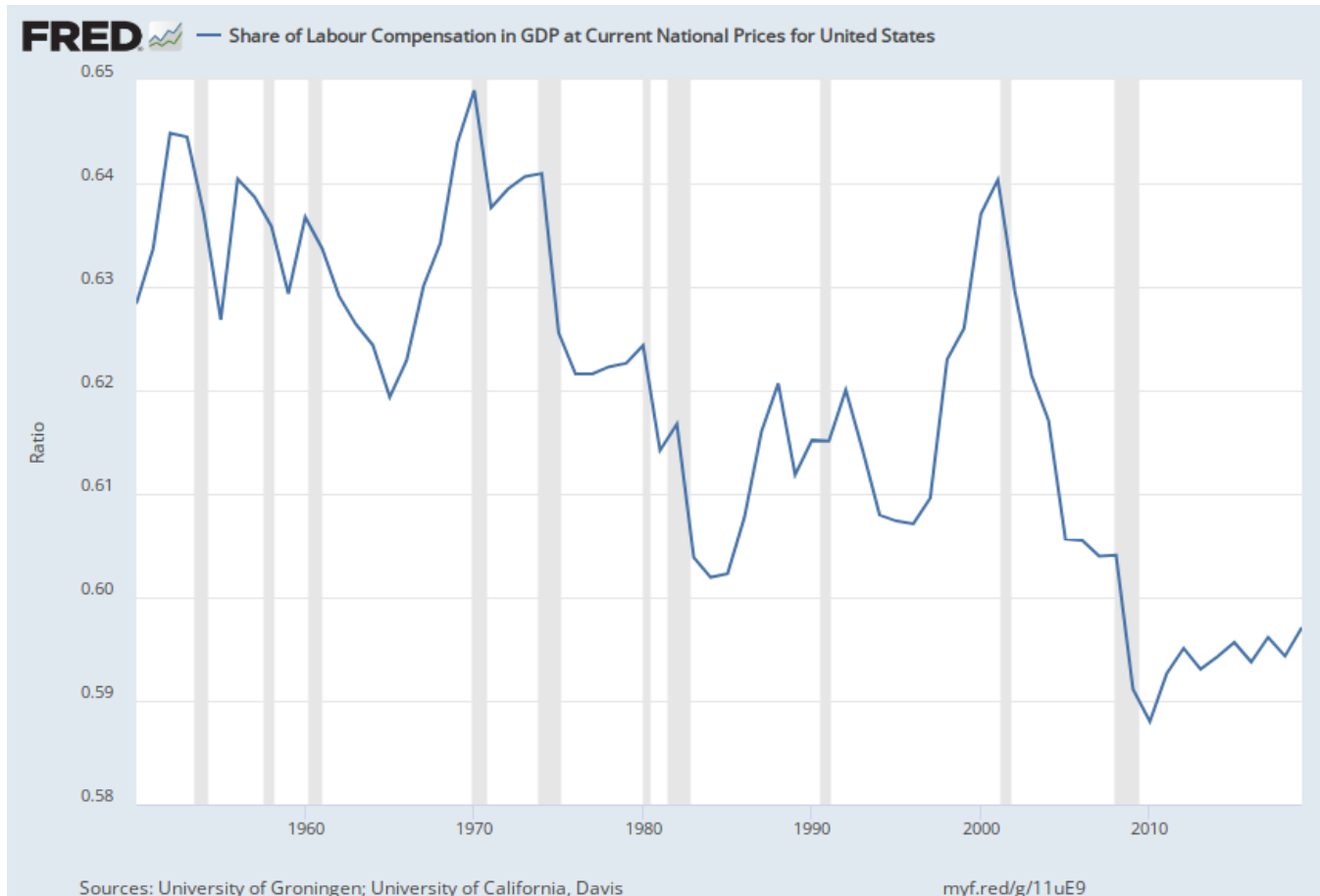
Labor income: €150

Capital income: €50

Indirect taxes: €0

$$\text{GDP} = \text{€150} + \text{€50} = \text{€200}$$

Labor share in GDP in the US since 1950



The labor share has been decreasing over time.
Conversely, the capital income/profit share has been increasing over time.

Housing in GDP

- Housing contributes notably to GDP, typically accounting for 15-18%, in two ways:
 - **Residential investment** (approximately 3-5% of GDP). This includes:
 - a. **New home construction**: Spending on newly built single-family homes, multi-family housing (apartments, condos), and mobile homes.
 - b. **Home improvements and renovations**: Expenditures on upgrades, remodeling, and major repairs to residential structures.
 - c. **Brokerage commissions and ownership transfer costs**: Fees associated with buying and selling homes, including real estate agent commissions and legal expenses.
 - **Consumption spending on housing services** (around 12-13% of GDP). This includes:
 - a. **Gross rents and utilities**: Payments made by renters for housing and related services.
 - b. **Owners imputed rents**: Estimate of the rental value of owner-occupied housing, including utilities.
- Including **owners imputed rent** in GDP has long been a standard practice in national income accounting. Were owners imputed rent not included, an increase in the homeownership rate would cause GDP to decline.

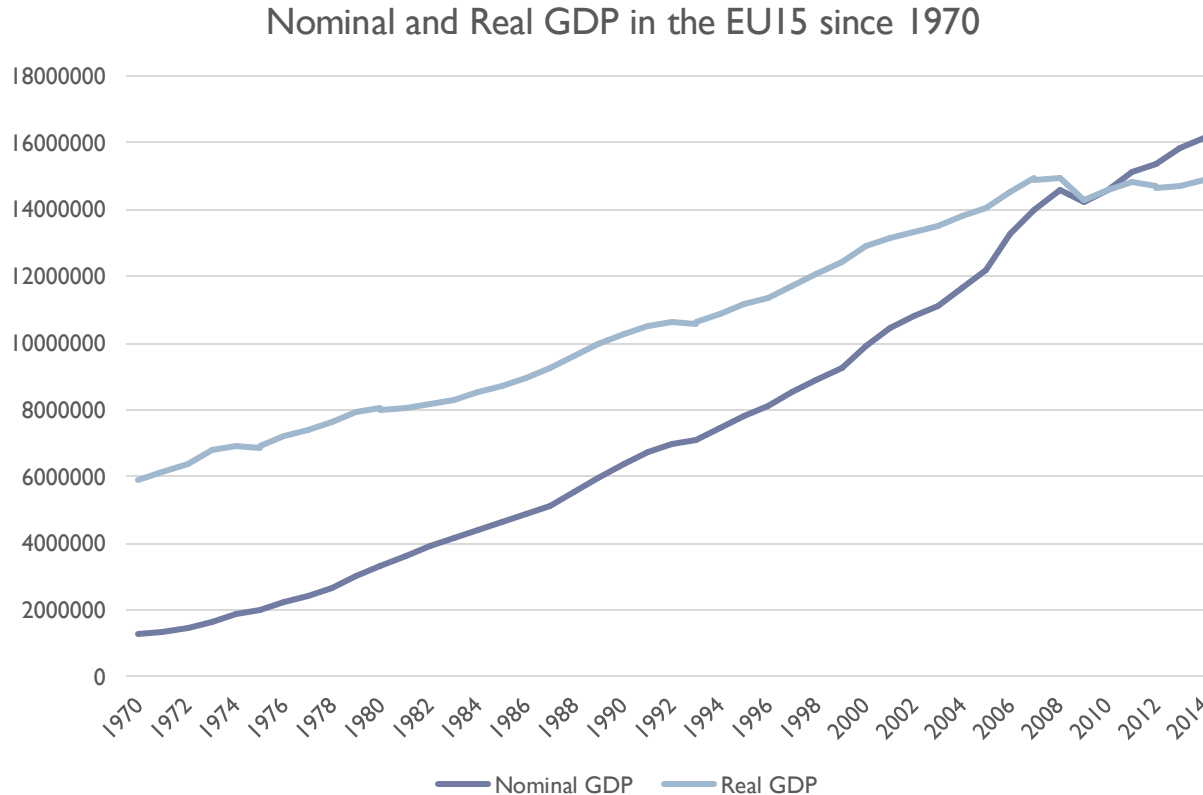
Nominal GDP

- **Nominal GDP** is the sum of the quantities of final goods produced multiplied by their **current prices**
- Nominal GDP increases over time because of 2 reasons
 - First, production of most goods increases over time
 - Second, prices of most goods also increase over time

Real GDP

- Real GDP is constructed as the sum of the quantities of final goods multiplied by constant prices (rather than current)
- Real GDP increases over time because production of most goods increases over time

Nominal and Real GDP



From 1970 to 2014, nominal GDP has increased by a factor of 13. Real GDP has increased by a factor of 2.5.

Measuring real GDP

- Simple in an economy with **one good**, say cars

Year	Quantity of cars	Price of cars	Nominal GDP	Real GDP (in 2005 prices)
2004	10	€20,000	€200,000	€240,000
2005	12	€24,000	€288,000	€288,000
2006	13	€26,000	€338,000	€312,000

- To construct real GDP, multiply the number of cars in each year by a **common price**
- Suppose we use the **price of cars in 2005** as the common price
- This approach gives us **real GDP in 2005 prices**
- **2005** is called the base year

Issue 1: Does measuring real GDP become more difficult in an economy with many final goods?

- With more than one good, the relative prices of the goods in the base year are used as natural weights

Example

Year	Cars		Books	
	Q^C	P^C	Q^B	P^B
2000	12	€24,000	10	€5,000
2010	13	€30,000	20	€2,000

Nominal GDP in 2000: $12 \times €24,000 + 10 \times €5,000 = €338,000$

Nominal GDP in 2010: $13 \times €30,000 + 20 \times €2,000 = €430,000$



Example (continued)

Year	Cars		Books	
	Q^C	P^C	Q^B	P^B
2000	12	€24,000	10	€5,000
2010	13	€30,000	20	€2,000

Real GDP-2000 (at 2000 prices): $12 \times €24,000 + 10 \times €5,000 = €338,000$

Real GDP-2010 (at 2000 prices): $13 \times €24,000 + 20 \times €5,000 = €412,000$

Real GDP-2000 (at 2010 prices): $12 \times €30,000 + 10 \times €2,000 = €380,000$

Real GDP-2010 (at 2010 prices): $13 \times €30,000 + 20 \times €2,000 = €430,000$

Issue 2: Will the level and rate of change of real GDP differ if we use prices from a different year?

- The level always changes with the base year
- The growth rate does not change with one good, but it does with many goods (since relative prices changes over time and hence depend on the base year)
- That it changes with many goods is not desirable. Several approaches: change the base year often; develop chained-type indexes

Recall - GDP: level versus growth rate

- Real GDP level in period t , Y_t
- Real GDP growth (or rate of change between $t-1$ and t):

$$(Y_t - Y_{t-1}) / Y_{t-1}$$

- Periods of **positive GDP growth** are called **expansions**
- Periods of **negative GDP growth** (at least 2 consecutive quarters) are called **recessions**

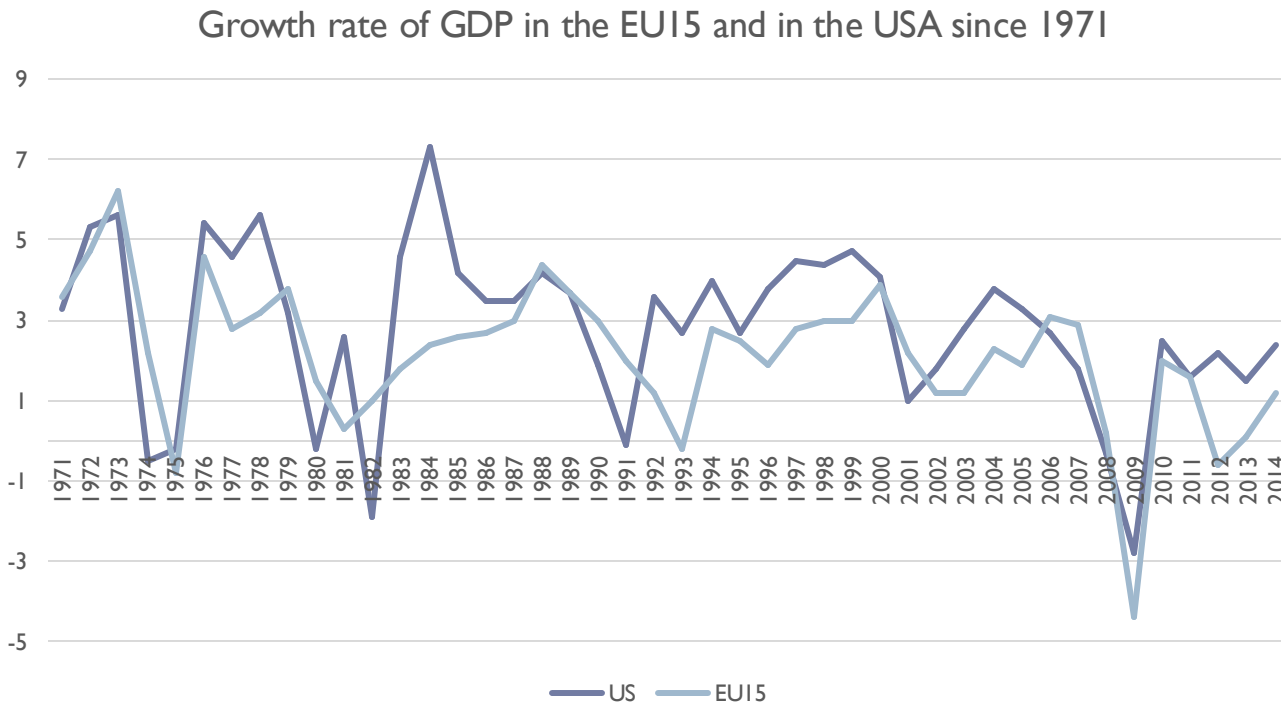
Back to our earlier example

- Simple economy with one good, say cars

Year	Quantity of cars	Price of cars	Nominal GDP	Real GDP (in 2005 prices)	Real GDP (in 2006 prices)
2004	10	€20,000	€200,000	€240,000	€260,000
2005	12	€24,000	€288,000	€288,000	€312,000
2006	13	€26,000	€338,000	€312,000	€338,000

- Real GDP growth using real GDP at 2005 prices
 - Growth from 2004 to 2005 = $(288,000 - 240,000) / 240,000 = 0,2 = 20\%$
 - Growth from 2005 to 2006 = $(312,000 - 288,000) / 288,000 = 0,083 = 8,3\%$
- Real GDP growth using real GDP at 2006 prices
 - Growth from 2004 to 2005 = $(312,000 - 260,000) / 260,000 = 0,2 = 20\%$
 - Growth from 2005 to 2006 = $(338,000 - 312,000) / 312,000 = 0,083 = 8,3\%$
- With one good, the growth rate of real GDP is unaffected by the base year

GDP growth rate



Since 1970, both the EU-15 and the US economy have gone through a series of expansions, interrupted by short recessions. The recession associated with the current crisis has been particularly deep.

Source: OECD *Economic Outlook* database

Inflation rate

- **General level of prices**

Average price of goods and services in the economy, P_t

- **Inflation rate**

Rate at which the general price level grows, π_t

$$\pi_t = (P_t - P_{t-1}) / P_{t-1}$$

Inflation and deflation

- **Inflation**

Raise in the price level; corresponds to a positive inflation rate

- **Deflation**

Decline in the price level; corresponds to a negative inflation rate

Price level

- In an economy with one final good it is straightforward to measure the price level; with many final goods it is more difficult
- Economists use 2 definitions of the aggregate price level, 2 price indexes
 - GDP deflator
 - Consumer price index
- The level of prices has no economic interpretation (index number); the rate of change has a clear interpretation (inflation)

GDP deflator

- **GDP deflator, P_t**

GDP deflator in t is defined as the ratio of nominal GDP to real GDP in t

$$\text{€}Y_t = P_t Y_t$$

$$\text{Nominal GDP} = \text{GDP deflator} \times \text{Real GDP}$$

$$P_t = \text{€}Y_t / Y_t$$

$$\text{GDP deflator} = \text{Nominal GDP} / \text{Real GDP}$$

- **Index number**

GDP deflator is an index number, set equal to 1 in the base year

Recall: Example

Year	Cars		Books	
	Q^C	P^C	Q^B	P^B
2000	12	€24,000	10	€5,000
2010	13	€30,000	20	€2,000

Nominal GDP in 2000: $12 \times €24,000 + 10 \times €5,000 = €338,000$

Nominal GDP in 2010: $13 \times €30,000 + 20 \times €2,000 = €430,000$

Recall: Example (continued)

Year	Cars		Books	
	Q^C	P^C	Q^B	P^B
2000	12	€24,000	10	€5,000
2010	13	€30,000	20	€2,000

Real GDP-2000 (at 2000 prices): $12 \times €24,000 + 10 \times €5,000 = €338,000$

Real GDP-2010 (at 2000 prices): $13 \times €24,000 + 20 \times €5,000 = €412,000$

Example (continued)

Deflator-2000 (at 2000 prices):

$$P_{2000} = \frac{\text{€}Y_{2000}}{Y_{2000}} = \frac{338,000}{338,000} = 1$$

Deflator-2010 (at 2000 prices):

$$P_{2010} = \frac{\text{€}Y_{2010}}{Y_{2010}} = \frac{430,000}{412,000} = 1.044$$

Inflation rate:

$$\pi_{2000-2010} = \frac{P_{2010} - P_{2000}}{P_{2000}} = \frac{1.044 - 1}{1} = 0.044$$

Example (continued)

- Does the inflation rate depend on the base year chosen? Yes
- Does the growth rate of real GDP depend on the base year chosen? Yes

Year	2000	2010	Growth
Nominal GDP	338,000	430,000	0.272
Real GDP (at 2000 prices)	338,000	412,000	0.219
Real GDP (at 2010 prices)	380,000	430,000	0.132
Deflator (at 2000 prices)	1.000	1.044	0.044
Deflator (at 2010 prices)	0.889	1.000	0.125

- Note that the rate of growth of nominal GDP is approximately equal to the rate of growth of real GDP plus the rate of inflation

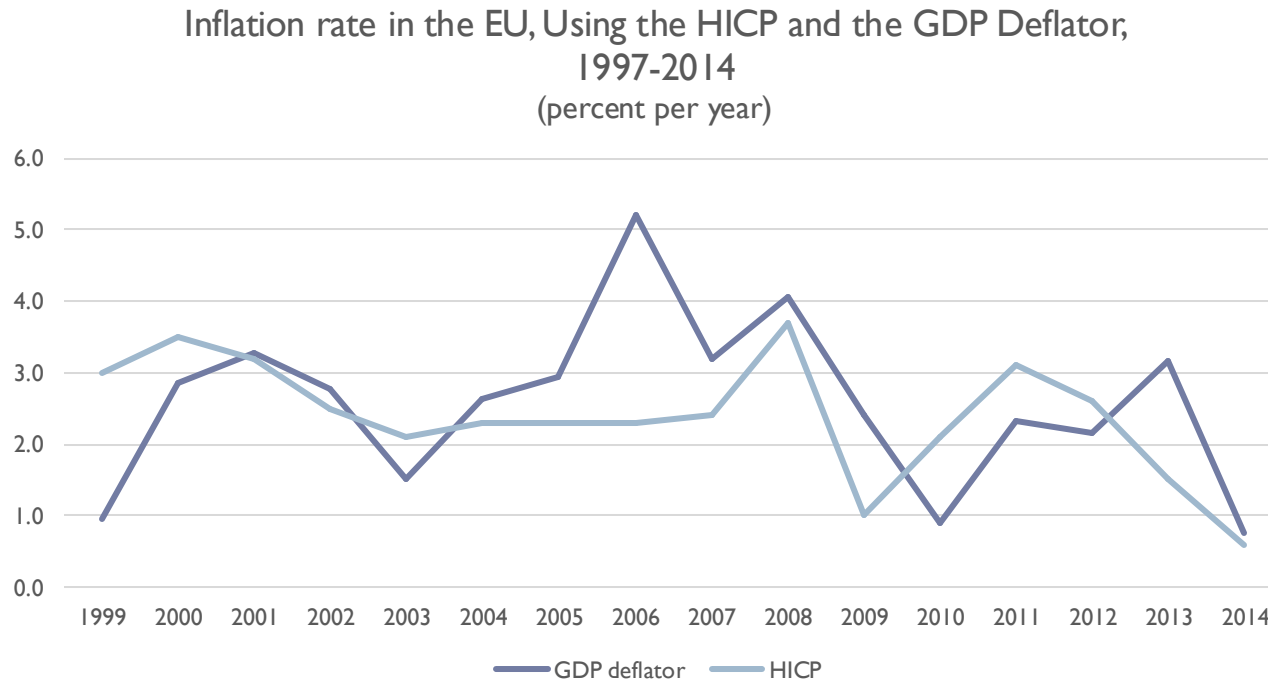
The Consumer Price Index (CPI)

- The GDP deflator measures the average price of output, i.e., of the final goods produced in the economy
- The consumer price index, or CPI, measures the average price of consumption, i.e., of the goods consumed by the households
- The CPI gives the cost in dollars of a specific list of goods and services over time, which attempts to represent the consumption basket of a typical urban consumer
- The CPI measures the cost of living

The set of goods produced in the economy is not the same as the set of goods purchased by consumers, for two reasons:

- Some of the goods are sold to firms, to the government or to foreigners
- Some of the goods are not produced domestically but are imported from abroad

Two inflation rates in the EA, 1999-2014



The inflation rates, computed using either the HICP or the GDP deflator, are largely similar.

Why do economists care about inflation?

Pure inflation: proportional increase in all prices and wages

- All relative prices unaffected
- Real wages for example unaffected

Economists care about inflation for two reasons:

- During periods of inflation, not all prices and wages rise proportionately, inflation affects relative prices and income distribution (no pure inflation)
- Inflation leads to other distortions due to uncertainty, prices that are fixed by law or regulation, and interaction with taxation

Unemployed workers

Thousands of unemployed people wait outside the State Labor Bureau in New York City to register for federal relief jobs in 1933

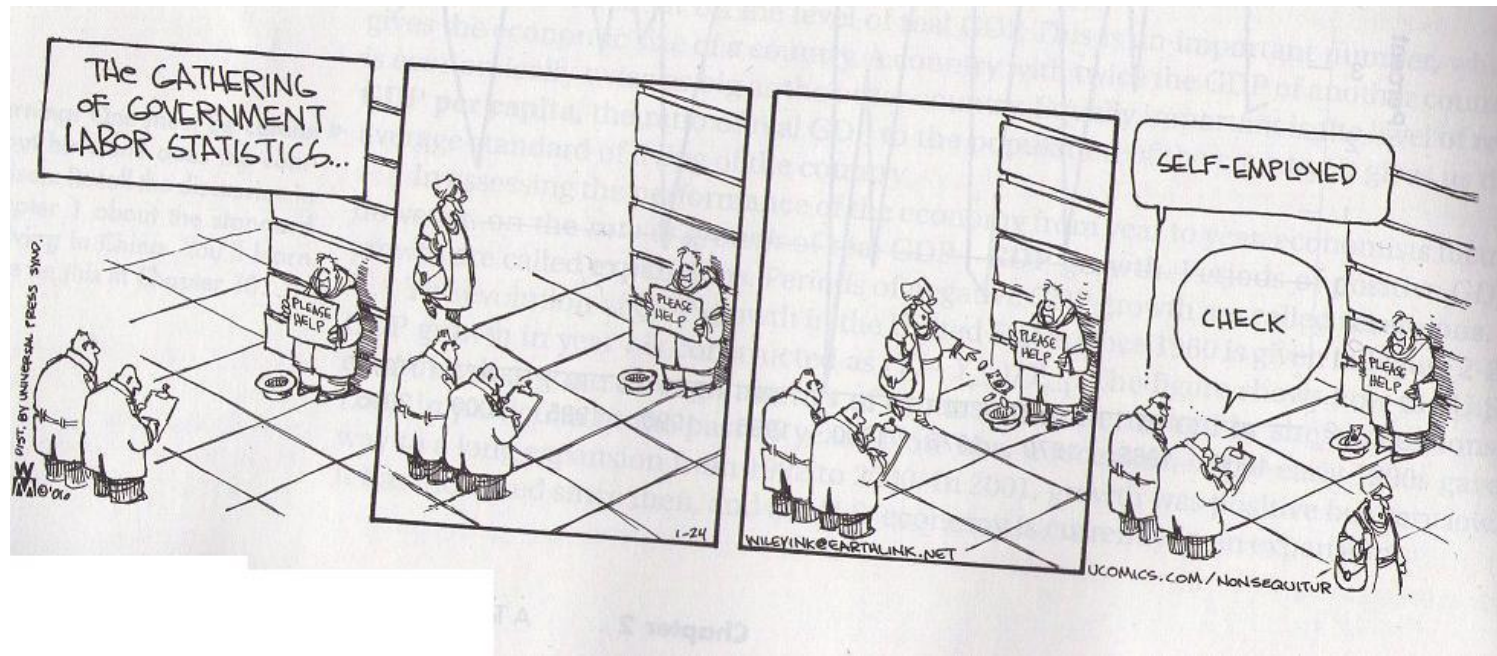


People who don't have a job but are looking for one

Measuring unemployment

- Measuring whether a person is employed is straightforward; measuring whether a person is unemployed is less obvious
- In past decades, only available source to measure unemployment was the [number of people registered at unemployment offices](#)
- This was a poor measure of unemployment: it depends on the generosity of unemployment benefit systems and on the exhaustion of benefits
- Most countries now rely on [Labor Force Surveys \(LFS\)](#): interviews of representative samples of individuals

Discouraged workers



People without jobs who give up looking for work

- **Working age population**

Number of people who can work, **P**

- **Employment**

Number of people who have a job, **N**

- **Unemployment**

Number of people who don't have a job but are looking for one, **U**

- **Labor force**

Sum of employment and unemployment, **L**

$$L = N + U$$

Labor force = Employment + Unemployment

$$P = L + \text{Out of labor force}$$

Working age population = Labor force + Out of the labor force

Unemployment rate

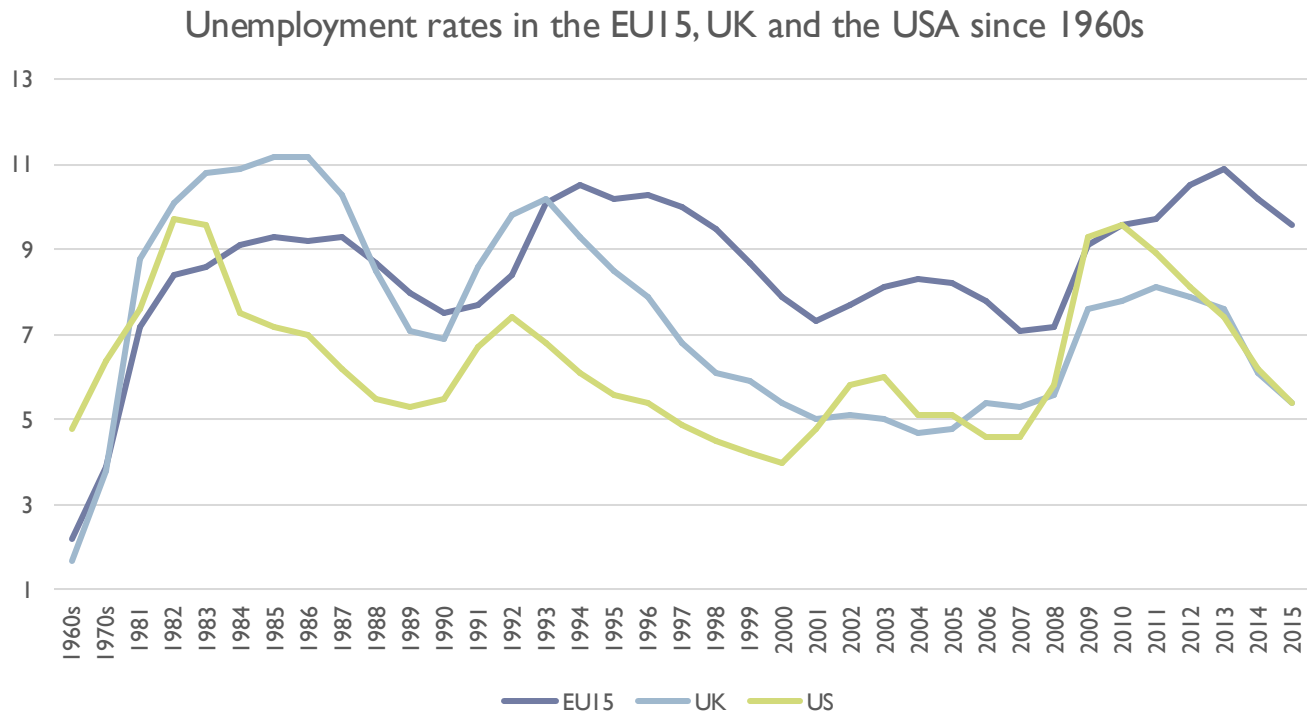
Ratio of unemployed people to the number of people in the labor force, u

$$u = U / L$$

Example: In 2008 in the US 144.4 million people were employed, 7.0 million people were unemployed, so the unemployment rate was $7.0 / (144.4 + 7.0) \times 100 = 7.1\%$

Question: what would happen to the unemployment rate if all workers without a job gave up looking for one?

Unemployment rate



Since the 1960s, the unemployment rate has fluctuated between less than 2% and more than 11% both in Europe and in the US, going down during expansions and going up during recessions. The effect of the crisis is highly evident, with the unemployment rate close to 10% in the EU15.

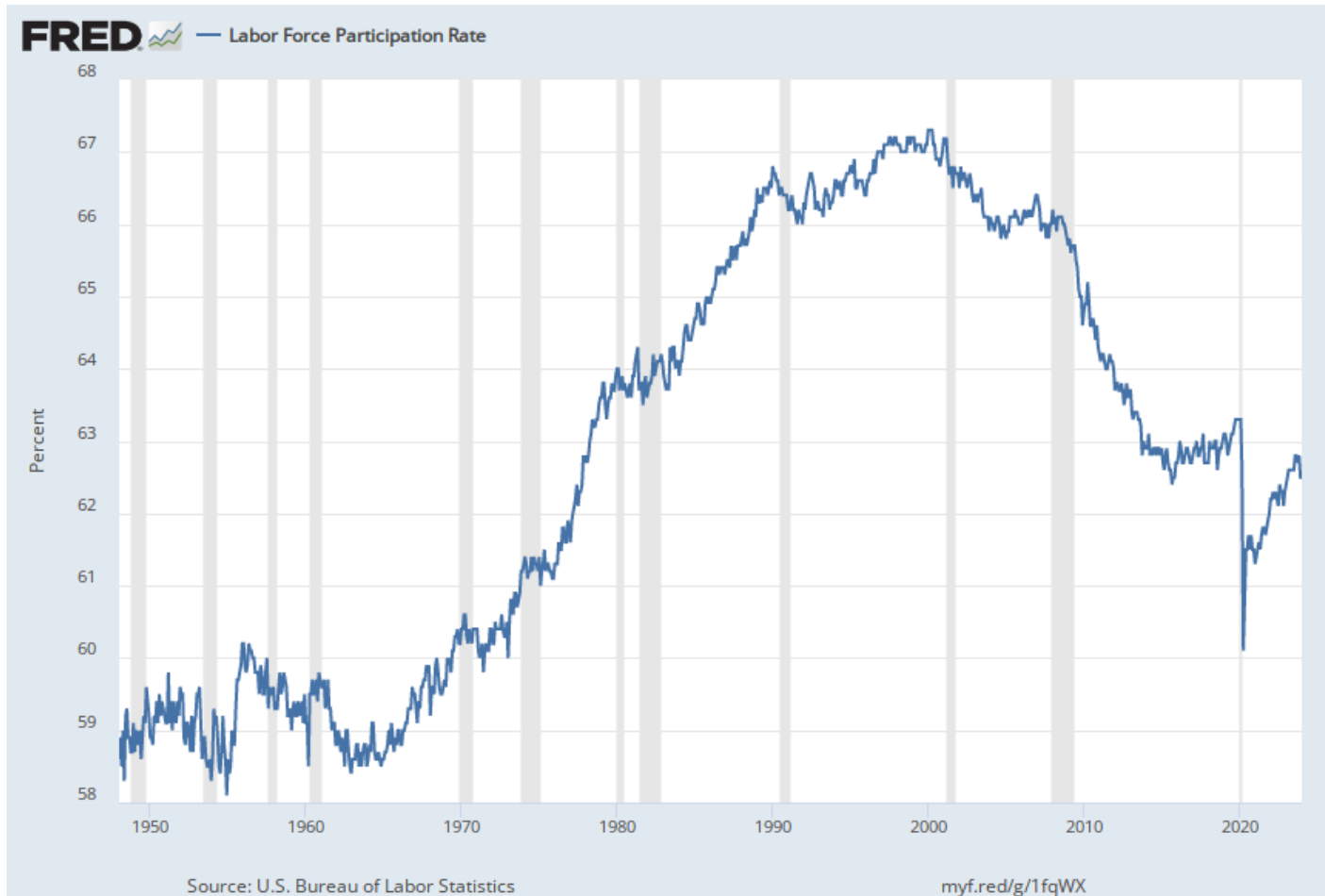
Participation rate

Ratio of people in the labor force to the working age population

$$\begin{aligned}\text{participation rate} &= L / P \\ &= (N + U) / (N + U + \text{Out of L})\end{aligned}$$

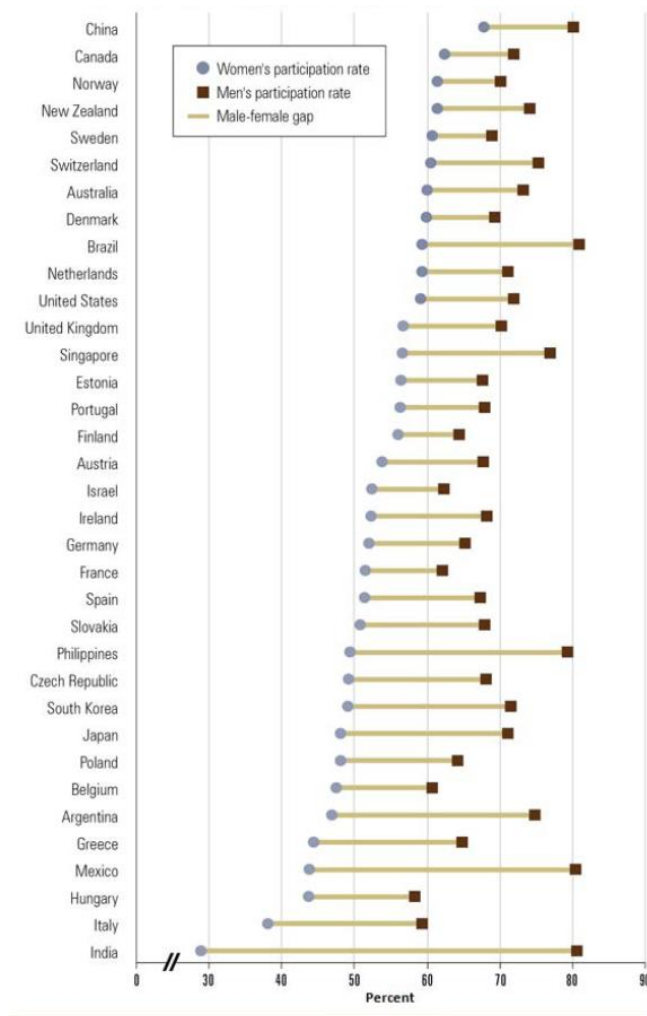
Question: what happens to the unemployment rate if the participation rate decreases because some unemployed leave the labor force?

Participation rate in the US since 1948



Participation rate

Labor force participation rates by sex, selected countries, 2010



Source: US Bureau of Labor Statistics and International Labor Office

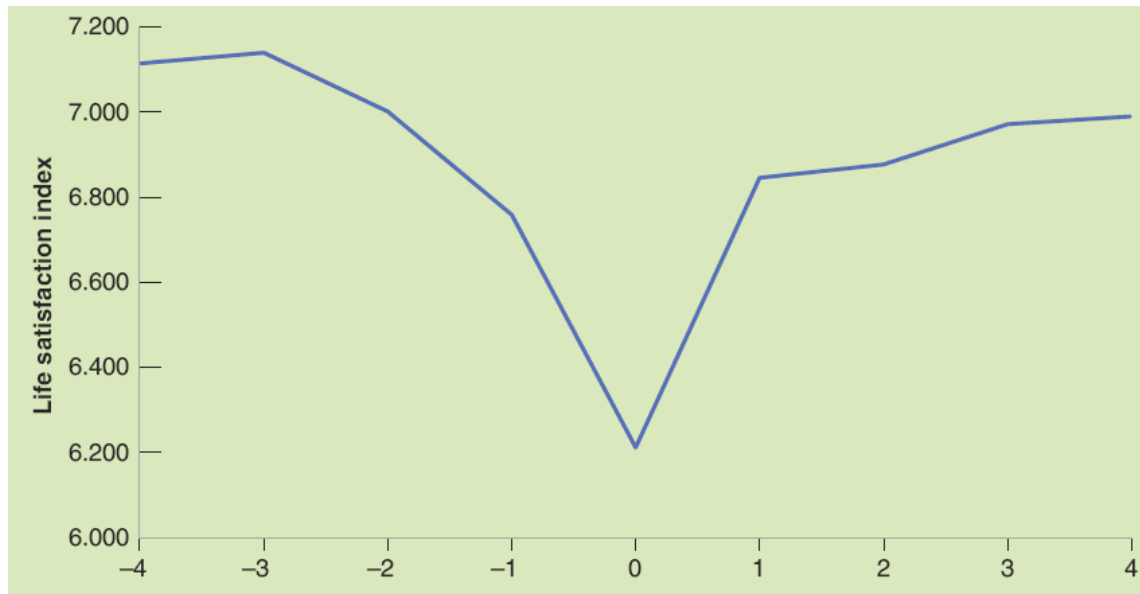
Why do economists care about unemployment?

Economists care about unemployment for two reasons:

- Because of its direct effects on the welfare of the unemployed, especially the long-term unemployed
- Because it provides a signal that the economy may not be using some of its resources efficiently

Effects of unemployment on happiness

The figure plots the average life satisfaction (happiness) for those individuals who were unemployed for 1 year and employed in the 4 years before and the 4 years after. Year 0 is the year of unemployment



Source: Winkelmann (2014)

Results suggest that (1) becoming unemployed leads to a large decrease in happiness, (2) happiness declines before the actual unemployment spell, and (3) happiness does not fully recover even four years later

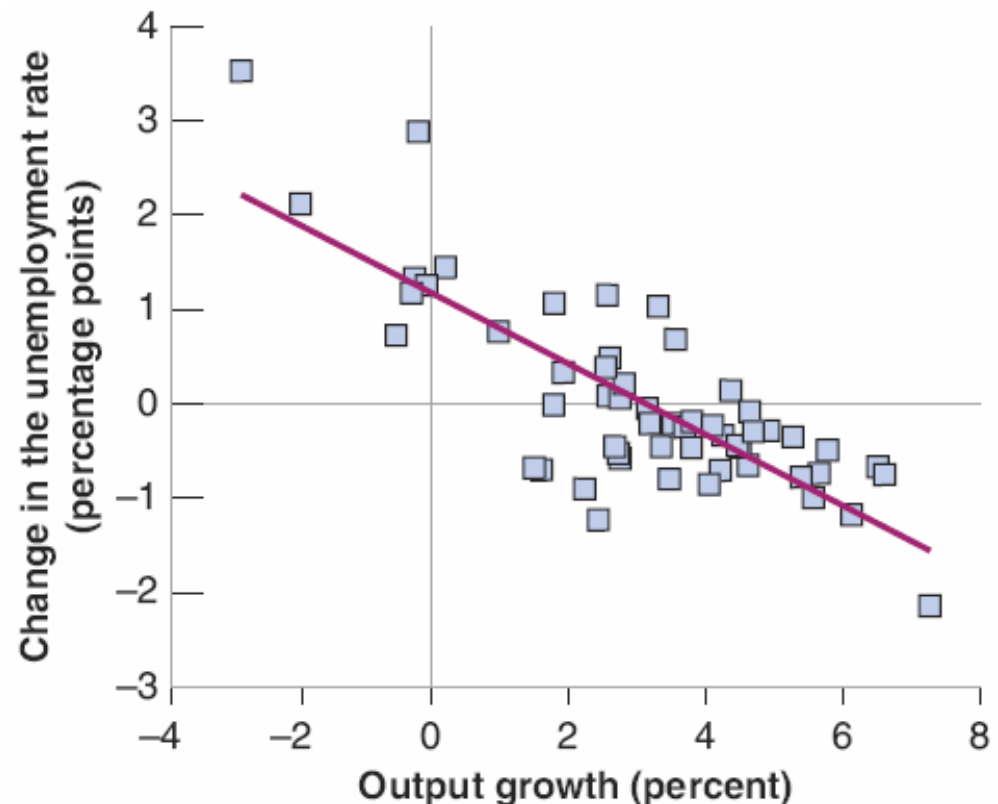
Okun's law

Changes in the Unemployment Rate versus Growth in the US, 1960-2014

Output growth higher than usual associated with a reduction in unemployment rate

Output growth lower than usual associated with an increase in unemployment rate

- **Okun's law** is a relation first examined by Arthur Okun
- The line that best fits the points is downward sloping
- The slope of the line is -0.4, which implies that, on average, an increase in the growth rate of 1% decreases the unemployment rate by -0.4%
- The line crosses the horizontal axis where output growth is 3%, meaning that it takes a growth rate of 3% to keep unemployment constant



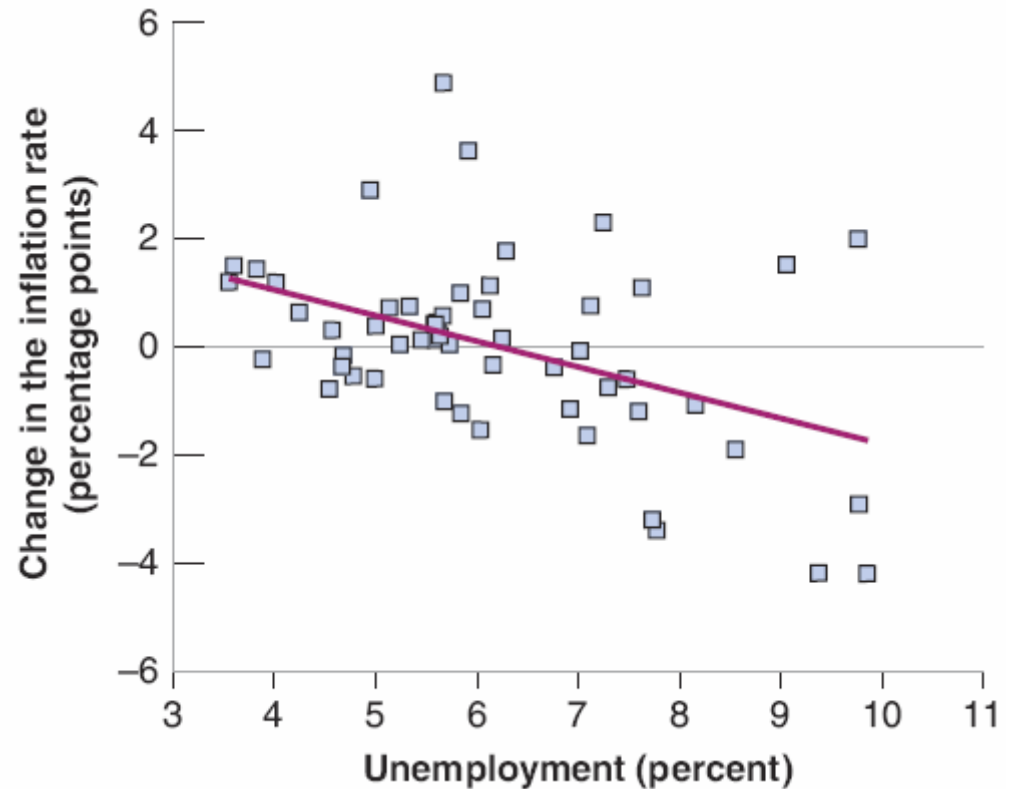
Phillips curve

Changes in the Inflation Rate versus the Unemployment rate in the US, 1960-2014

Unemployment rate higher than usual leads to an increase in the inflation rate

Unemployment rate lower than usual leads to a decrease in the inflation rate

- The **Phillips' curve** is a relation first examined by A. W. Phillips
- The line that best fits the points is downward sloping, although the fit is not as good as it was for Okun's law
- The line crosses the horizontal axis where the unemployment rate is equal to about 6%, meaning that inflation typically increased when unemployment was below 6% and decreased when unemployment was above 6%



Short run, medium run and long run

Level of aggregate output in an economy is determined by:

- Demand in the short run (a few years)
- Level of technology, the capital stock and the labor force in the medium run (a decade or so)
- Factors such as education, research, saving and the quality of government in the long run (half a century or more)

A Tour of the Book

